



Establishing community reference samples, data and call sets for benchmarking cancer mutation detection using whole-genome sequencing

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The lack of samples for generating standardized DNA datasets for setting up a sequencing pipeline or benchmarking the performance of different algorithms limits the implementation and uptake of cancer genomics. Here, we describe reference call sets obtained from paired tumor-normal genomic DNA (gDNA) samples derived from a breast cancer cell line—which is highly heterogeneous, with an aneuploid genome, and enriched in somatic alterations—and a matched lymphoblastoid cell line. We partially validated both somatic mutations and germline variants in these call sets via whole-exome sequencing (WES) with different sequencing platforms and targeted sequencing with >2,000-fold coverage, spanning 82% of genomic regions with high confidence. Although the gDNA reference samples are not representative of primary cancer cells from a clinical sample, when setting up a sequencing pipeline, they not only minimize potential biases from technologies, assays and informatics but also provide a unique resource for benchmarking ‘tumor-only’ or ‘matched tumor-normal’ analyses.

Accurate somatic mutation detection is essential for cancer genomics^{1–4} and precision cancer medicine^{5,6}. Despite rapid advances in sequencing technologies, it remains challenging to detect somatic mutations accurately using next-generation sequencing (NGS). It is difficult to obtain consistent and concordant somatic mutation calls from individual platforms or pipelines^{7–9}, hampering the development of personalized therapies. In addition, quality control of somatic mutation pipelines is often insufficient due to the lack of well-validated and publicly available reference samples and reference datasets. Therefore, paired tumor-normal reference samples with high-confidence somatic mutation calls are desirable and urgently needed¹⁰. As more sequencing technologies and bioinformatics pipelines are used to identify clinically actionable somatic mutations, the need grows even stronger⁵.

Although reference samples and call sets have been established for benchmarking germline variant calls by the Genome in a Bottle

(GIAB) Consortium^{11,12}, no such resource exists for benchmarking somatic variant calls. Thus, reference samples for germline variant calls are often used unwillingly to validate somatic mutation pipelines^{13–15}. However, accurate detection of somatic mutations is much more challenging due to variable variant allele frequency (VAF), inter- and intratumor heterogeneity, prevalent copy number alterations (CNAs) and complex chromosomal rearrangements. Although gene-specific reference samples are available^{16,17} (<https://www.horizondiscovery.com/reference-standards/type/oncospan>), they are unsuitable to benchmark sequencing pipelines applied to the whole genome. Simulated sequence alignment data have been used to benchmark calling algorithms of single-nucleotide variants¹⁸ (SNVs) and structural variations¹⁹, but they are not feasible to benchmark sequencing technologies, sequencing protocols and alignment algorithms. The International Cancer Genome Consortium (ICGC) argued that real, not simulated, somatic mutations are preferred, and

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Table 1 | Data from multiple NGS platforms were obtained to detect and confirm germline and somatic variants in HCC1395 and HCC1395BL cell lines

Technology		Platforms	Sequencing	Number of reads (coverage)	
				HCC1395	HCC1395BL
Discovery	WGS	HiSeq	6 centers 12 libraries	21 billion (750×)	21 billion (750×)
		NovaSeq	1 center 9 libraries	13 billion (400×)	13 billion (400×)
Validation	WGS-tumor content	HiSeq	1 center 3 libraries	9.2 billion (350×)	9 billion (350×)
	Tumor-normal mix	HiSeq	1 center 15 libraries	46 billion (1,500×)	
	WGS	PacBio	1 center 1 library	20 million (40×)	20 million (40×)
	WES	HiSeq	6 centers 12 libraries	3 billion (2,500×)	3 billion (2,500×)
		Ion Torrent	1 center 1 library	67 million (34×)	82 million (47×)
	AmpliSeq	MiSeq	1 center 1 library	25 million (2,900×)	22 million (3,300×)
	Microarray	AffyChip CytoScan HD	1 center 1 library	2.1 million probes	
	Single-cell sequencing	10x Genomics	1 center 1 library	1,465 cells	983 cells

ICGC provided two whole-genome sequencing (WGS) benchmark datasets (from a chronic lymphocytic leukemia (CLL) case and a medulloblastoma (MB) case²⁰). Similarly, a somatic WGS reference call set of a metastatic melanoma cell line has been provided²¹. Somatic mutations called in those two reference samples have not been successfully validated by orthogonal sequencing technologies. ICGC tried MiSeq and Ion Torrent for the CLL case, but the validation rate was low due to technical issues²⁰. A recent landscape survey of current somatic reference samples by the Medical Devices Innovation Consortium did not identify any readily accessible tumor-normal reference samples for evaluating the somatic mutation calling accuracy on a whole-genome basis²².

To establish reference data and call sets for benchmarking somatic mutation calling, we extracted gDNA samples from a triple-negative breast cancer (TNBC) cell line (HCC1395) and a B lymphocyte-derived normal cell line (HCC1395BL) from the same donor from the American Type Culture Collection (ATCC). Unlike reference call sets derived from CLL and MB tumors with low mutation burden and very limited structural changes, the HCC1395 cell line is rich in genomic alterations (~40,000 SNVs, ~2,000 small insertions and deletions (indels), CNAs in ~56% of the genome and >256 complex genomic rearrangements²³, an aneuploid genome and BRCAness²⁴) and has been characterized previously using cytogenetic analysis²⁵ and array-based comparative genomic hybridization²⁶. Moreover, we preferred HCC1395 over a normal cell line with engineered mutations^{27–29} or synthetic DNA spike-in³⁰ because it is difficult, if not impossible, to engineer a whole range of somatic mutations into host chromosomes to mimic the complexity and heterogeneity of a cancer genome.

Readers should note, however, that the genome of the HCC1395 cell line does not necessarily represent TNBC cancer genomes accurately. Unlike the 1000 Genomes Project³¹, which served as a detailed catalog of common human germline variants, our reference call sets are not designed to be a detailed catalog of somatic mutations of breast cancer. Rather, they include a wide range of mutations that occur naturally in vivo and later in cell cultures and, thus, are suitable to benchmark, develop and refine protocols and tools for somatic mutation detection. In addition, our reference call sets were developed with gDNA extracted from fresh cells and sequenced by Illumina short-read technology. Thus, they are not suitable for benchmarking different sequencing technologies or applications using different sample types, such as formalin-fixed paraffin-embedded (FFPE) samples or circulating tumor DNA (ctDNA).

We sequenced the whole genomes of the paired tumor-normal cell lines using WGS (at 1,500× coverage) across seven sequencing

centers. The sequencing reads were aligned, and somatic mutations were called by various bioinformatics pipelines. Thus, we minimized biases specific to the sequencing platforms and centers, or the bioinformatics algorithms, and created high-confidence mutation calls across the whole genome of the HCC1395 cell line, namely the ‘reference call set’. A subset of these mutation calls was validated by targeted sequencing (at 2,000× coverage), WES using HiSeq (at 2,500× coverage) and Ion Torrent (at 34× coverage) and long-read WGS by PacBio Sequel (at 40× coverage). In addition, we inferred subclones and heterogeneity of the HCC1395 cell line with bulk DNA sequencing. The results were confirmed by single-cell DNA sequencing analysis. Finally, we provided high-confidence germline and somatic call sets with clinically relevant annotations.

The established call sets have at least two advantages over the others: (1) the whole genomes of reference samples were sequenced with deeper coverage at a total of 1,500× and were validated by orthogonal sequencing platforms; consequently, high-confidence clonal and subclonal somatic mutations were called and validated; and (2) somatic mutations called from 378 datasets (21 sequencing replicates by three aligners and by six somatic mutation callers) were consolidated by two state-of-the-art machine learning-based somatic mutation classifiers (SomaticSeq³² and NeuSomatic³³) to construct a high-confidence somatic reference call set, which alleviated calling errors specific to sequencing platform, sequencing site or bioinformatics algorithm.

Results

Study design with multiple sequencing platforms and various variant calling pipelines. gDNA was extracted from each of the matched cell lines (HCC1395 and HCC1395BL) in single batches for the study. To establish the high-confidence call sets for these batches of gDNA, we generated high-coverage short-read WGS data using multiple platforms at seven sequencing centers. We performed long-read WGS (with PacBio), WES (with HiSeq and Ion Torrent) and AmpliSeq (with MiSeq) to confirm our findings and used CytoScan microarray and 10x Genomics single-cell copy number variation (CNV) analysis to uncover the cytogenetic properties and heterogeneity of the two cell lines (Table 1).

The initial somatic mutation call sets were obtained from 21 replicates of HCC1395 and HCC1395BL cells with three aligners (BWA-MEM³⁴, Bowtie2 (ref. ³⁵) and NovoAlign (<http://www.novocraft.com/products/novoalign/>)) and six mutation callers (MuTect2 (ref. ¹³), SomaticSniper³⁶, VarDict³⁷, MuSE³⁸, Strelka2³⁹ and TNscope⁴⁰). We used SomaticSeq³² and NeuSomatic³³ to score each mutation call based on evidence collected from the 378 call sets.

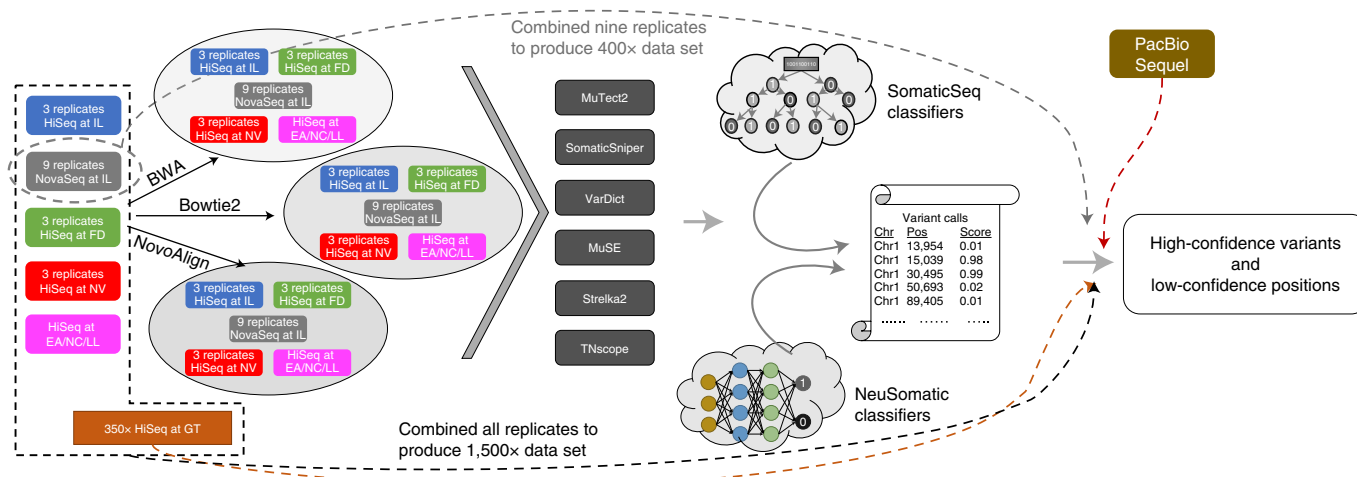


Fig. 1 | Schematic of the bioinformatics pipelines used to define the confidence levels of the somatic mutation call set (see Methods for details).

Twenty-one sequencing replicates for the tumor (HCC1395) and normal (HCC1395BL) gDNA samples were performed at six sequencing centers. They were grouped into five groups shown as colored squares on the far left. The sequencing platforms used were HiSeq and NovaSeq, and the sequencing experiments were performed at Illumina (IL), Fudan University (FD), Novartis (NV), European Infrastructure for Translational Medicine (EA), National Cancer Institute (NC) and Loma Linda University (LL). Each of the 21 sequencing datasets was aligned with three aligners to create a total of 63 pairs of tumor-normal BAM files. For each of the 63 tumor-normal BAM files, we ran six somatic mutation callers (MuTect2, SomaticSniper, VarDict, MuSE, Strelka2 and TNscope) followed by SomaticSeq and NeuSomatic machine learning classifiers using models built specifically for these datasets. Initial tier was assigned to each variant call based on how consistently the variant was classified as a somatic mutation across different sequencing centers and aligners. Then, to rescue low-VAF variants into the reference call set, we ran the same variant-calling pipeline on two higher-depth datasets: an independent 350x replicate sequenced on a HiSeq at Genentech (GT) and a 400x replicate by combining nine NovaSeq replicates at Illumina (IL). High-confidence calls from those two datasets were used to promote less-reproducible low-VAF variants from the 21 sequencing replicates into the reference call set. Then, we combined all of our short-read sequencing data into a pair of 1,500x tumor-normal and used it to rescue additional low-VAF variants into the reference call set. Finally, we cross-referenced our Illumina short-read-based high-confidence calls with PacBio long-read sequencing data and removed a small number of calls that were inconsistent with each other. The high-confidence calls (labeled PASS) were considered 'true' somatic mutations, and genomic regions with low-confidence calls were removed from the high-confidence regions. Chr, chromosome; Pos, position.

In addition, two pooled datasets with 400x coverage from NovaSeq data of nine replicates and 350x coverage from the WGS-tumor content replicates were used to improve mutation calls with VAFs of <15%. Finally, we used NeuSomatic³³ to confirm and complement the mutation calls with all WGS data with 1,500x coverage (Fig. 1).

Definition of the somatic reference call set and high-confidence regions. We first combined call sets from the six callers for each of the 63 pairs of tumor-normal data (21 replicates by three aligners) and then used SomaticSeq to classify each variant call from each of the 63 pairs of tumor-normal data into PASS, REJECT or LowQual. All 63 SomaticSeq call sets were combined. Using cross-aligner and cross-sequencing center reproducibility, each variant was then assigned one of the four confidence levels: high confidence (HighConf), medium confidence (MedConf), low confidence (LowConf) and Unclassified. For low-VAF calls, three datasets (HiSeq dataset (350x coverage), NovaSeq dataset (400x coverage) and a combination of all WGS runs (1,500x coverage)) were used to salvage some LowConf and Unclassified calls as MedConf. A few HighConf calls that were inconsistent with the PacBio data were demoted to LowConf. The call set in its entirety was designated the 'all-inclusive set', which included LowConf and likely false-positive (Unclassified) calls. HighConf and MedConf calls in the all-inclusive set were grouped as the 'somatic reference call set' and treated as true positives (see Methods and Supplementary Information for details). A breakdown of the four confidence levels in the all-inclusive set is shown in Fig. 2a.

Because NeuSomatic outperformed other callers in challenging situations, such as low-coverage, low-VAF and difficult genomic regions⁴¹, we applied NeuSomatic to each of the 63 call sets and

created mutation classifiers as SomaticSeq did. The classification results from the decision tree-based adaptive-boosting approach used by SomaticSeq and the convolutional neural network-based deep learning algorithm used by NeuSomatic were highly correlated (Pearson's $R = 0.997$) (Extended Data Fig. 1). We removed 526 SNVs and 60 indels that were discrepant between the two classifiers to improve precision and specificity of the somatic reference call set.

The 21 WGS runs used for the initial call set definition ranged from 50x to 100x coverage, and VAFs for most SNVs in the HighConf set were >5%. To improve sensitivity for low-VAF (<5%) variants, we pooled the short-read WGS sequencing data (1,500x coverage) to make NeuSomatic calls. We observed very high concordances for HighConf (99.6%) and MedConf (93.0%) SNVs between the initial call set and the call set from the 1,500x pooled data. Following manual inspection, most of the discordant calls (that is, classified as somatic mutations in the initial call set but not in the pooled dataset) were, indeed, likely true somatic mutations. They mostly consisted of dinucleotide/trinucleotide changes and low-VAF variants that were not confidently detected in the single 1,500x data. However, only 24.0% and 1.67% of LowConf and Unclassified SNVs in the initial call set, respectively, were called in the pooled dataset (Supplementary Table 1). In other words, LowConf and Unclassified SNVs were unlikely to be rediscovered in the pooled 1,500x dataset. Most of the Unclassified calls in the pooled call sets were complex variants (that is, more complex than simple dinucleotide/trinucleotide changes). In addition, HighConf and MedConf SNVs had high validation rates by AmpliSeq (Fig. 2b). Therefore, we used a conservative approach to allow low-VAF and low-signal variants into the somatic reference call set (that is, if they were supported by NeuSomatic in the 1,500x

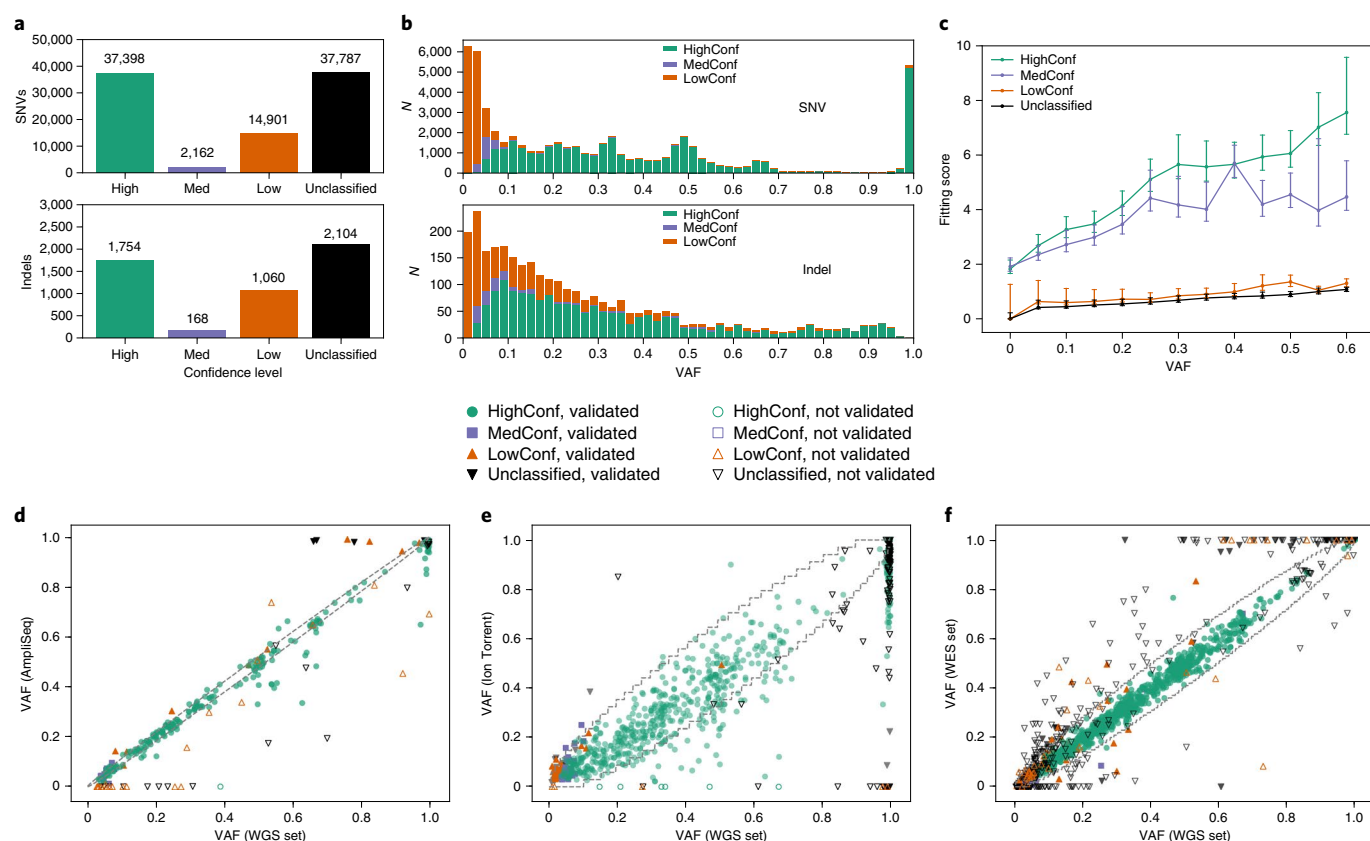


Fig. 2 | Definition and validation of the somatic mutation reference call set. **a**, Breakdown of the somatic variant calls within the consensus callable regions based on the four labels HighConf, MedConf, LowConf and Unclassified. Variant calls labeled HighConf and MedConf are grouped into the reference call set; genomic positions with LowConf and Unclassified calls are removed from the high-confidence regions. **b**, Histogram of VAFs of the somatic variant calls. **c**, Average tumor purity fitting scores with 95% confidence intervals for the VAF of each SNV across the four different confidence levels versus the observed VAF in the tumor-normal titration series. The formula for fitting scores is described in equation 1 (see Methods for details). **d**, Scatter plot of VAFs observed in 21 WGS datasets versus an AmpliSeq targeted sequencing dataset. Solid shapes represent variants that were validated. Open shapes represent variants that were not validated. Sticks represent uninterpretable validation data. The diagonal dashed lines represent the 95% binomial confidence interval of the observed VAF given the actual VAF calculated based on 2,000× depth for AmpliSeq. The figure shows a very high correlation between VAFs estimated from the WGS data and AmpliSeq data for HighConf calls (Pearson's $R=0.982$). Many Unclassified data points lie at the bottom, implying that these calls were not real mutations, despite the large number of apparent variant-supporting reads in the all-inclusive set data; x axis, VAFs calculated from the all-inclusive set; y axis, VAFs calculated from the AmpliSeq data. **e**, Scatter plot of VAFs observed in WGS datasets versus Ion Torrent WES. The 95% binomial confidence intervals were calculated based on 34× depth for Ion Torrent. Pearson's $R=0.930$ for HighConf calls. **f**, Scatter plot of VAFs observed in WGS datasets versus 12 repeats of WES on the HiSeq platform; y axis, median VAFs calculated based on 12 HiSeq WES replicates. The 95% binomial confidence intervals were calculated based on 150× depth for HiSeq WES. Pearson's $R=0.997$ for HighConf calls. In **d–f**, the colors indicate the confidence level of the variant calls, whereas the shapes indicate their validation status.

data in all three aligners (see Methods for details)). Consequently, 894 SNVs and 33 indels were ‘rescued’ into the somatic reference call set (Supplementary Data File 1).

It is well known that false positives are overwhelmingly enriched in genomic regions where alignment is challenging⁴². High GC content or low-complexity regions in the human genome cannot be covered adequately by current short-read technologies (Extended Data Fig. 2a). To obtain the callable regions in the reference genome, we ran GATK CallableLoci on each of the 63 BAM files to identify regions of low coverage (<10 reads), ultra-high coverage (more than eight times the mean coverage of the sample), mapping difficulty (mapping quality (MQ) <20) and poor-quality reads (base quality (BQ) score <20). From the consolidated CallableLoci results, we further removed the three chromosomal regions with normal loss of heterozygosity (LOH) and genomic regions with LowConf and Unclassified calls and created a list of high-confidence regions for the somatic reference call set, containing a total of 2.48 billion base pairs (bp) (Extended Data Fig. 2b). The derived regions were

callable with short-read sequencing technologies (that is, Illumina platforms). Variant calls outside the high-confidence regions were labeled as ‘NonCallable’ in the all-inclusive set.

Validation of mutations in the somatic reference call set. To confirm HighConf and MedConf calls, we mixed tumor HCC1395 gDNA with normal HCC1395BL gDNA at different ratios to generate a range of admixtures that represented tumor purity levels of 100%, 75%, 50%, 20%, 10%, 5% and 0% and performed WGS with 350× coverage. We calculated the tumor purity fitting scores (described in the Methods, equation 1) for measuring the concordance between the expected VAF and the observed VAF across tumor purity levels (Fig. 2c). For real somatic mutations in copy number-neutral regions, the observed VAFs should scale linearly with the tumor fraction in the tumor–normal titration series and show a higher score but not for sequencing artifacts or germline variants. The fitting scores for HighConf and MedConf calls were much higher than LowConf and Unclassified calls across all

Table 2 | Validation of somatic SNVs and indels with different confidence labels by three different methods

Validation platform	Variant type	Category	Total number	Fraction interpretable	Validation rate (interpretable) ^a	Validation rate (total) ^b
AmpliSeq deep sequencing	SNV	HighConf	251	(245/251) 97.6%	(244/245) 99.6%	(244/251) 97.2%
		MedConf	18	(18/18) 100.0%	(18/18) 100.0%	(18/18) 100%
		LowConf	51	(40/51) 78.4%	(19/40) 47.5%	(19/51) 37.3%
		Unclassified	35	(22/35) 62.9%	(4/22) 18.2%	(4/35) 11.4%
	Indel	HighConf	18	(18/18) 100.0%	(17/18) 94.4%	(17/18) 94.4%
		MedConf	1	(1/1) 100.0%	(1/1) 100.0%	(1/1) 100.0%
		LowConf	0	(0/0) NA ^c	(0/0) NA	(0/0) NA
		Unclassified	2	(1/2) 50.0%	(0/1) 0.0%	(0/2) 0.0%
Ion Torrent WES	SNV	HighConf	713	(636/713) 89.2%	(630/636) 99.1%	(630/713) 88.4%
		MedConf	50	(27/50) 54.0%	(27/27) 100.0%	(27/50) 54.0%
		LowConf	216	(38/216) 17.6%	(33/38) 86.8%	(33/216) 15.3%
		Unclassified	519	(110/519) 21.2%	(17/110) 15.5%	(17/519) 3.3%
	Indel	HighConf	41	(30/41) 73.2%	(26/30) 86.7%	(26/41) 63.4%
		MedConf	3	(1/3) 33.3%	(1/1) 100.0%	(1/3) 33.3%
		LowConf	9	(2/9) 22.2%	(2/2) 100.0%	(2/9) 22.2%
		Unclassified	24	(3/24) 12.5%	(3/3) 100.0%	(3/24) 12.5%
HiSeq WES	SNV	HighConf	1,089	(1,088/1,089) 99.9%	(1,088/1,088) 100.0%	(1,088/1,089) 99.9%
		MedConf	78	(75/78) 96.2%	(75/75) 100.0%	(75/78) 96.2%
		LowConf	421	(243/421) 57.7%	(203/243) 83.5%	(203/421) 48.2%
		Unclassified	1,162	(538/1,162) 46.3%	(72/538) 13.4%	(72/1,162) 6.2%
	Indel	HighConf	57	(57/57) 100.0%	(56/57) 98.2%	(56/57) 98.2%
		MedConf	3	(3/3) 100.0%	(3/3) 100.0%	(3/3) 100%
		LowConf	15	(13/15) 86.7%	(10/13) 76.9%	(10/15) 33.3%
		Unclassified	49	(26/49) 53.1%	(10/26) 38.5%	(10/49) 20.4%
Total unique counts (validated on at least one platform)	SNV	HighConf and MedConf	1,428	(1,418/1,428) 99.3%	(1,417/1,418) 99.93%	(1,417/1,428) 99.2%
		LowConf and Unclassified	1,668	(853/1,668) 51.1%	(313/853) 36.69%	(313/1,668) 18.8%
	Indel	HighConf and MedConf	82	(80/82) 97.6%	(78/80) 97.50%	(78/82) 95.1%
		LowConf and Unclassified	67	(40/67) 59.7%	(20/40) 50.00%	(20/67) 29.9%

^aThe 'interpretable' validation rates only count variant positions that were determined as either 'validated' or 'not validated'. ^bThe 'total' validation rate counts uninterpretable genomic positions as 'not validated'; hence, it is a lower limit of validation rates. ^cNA, percentage not applicable.

VAF brackets, indicating that HighConf and MedConf calls contained notably more real somatic mutations than LowConf and Unclassified calls.

In addition, we randomly selected a number of mutation calls at HighConf, MedConf, LowConf and Unclassified in the all-inclusive set for validation by AmpliSeq with PCR enrichment in addition to Ion Torrent WES and HiSeq WES (SNVs, Fig. 2d–f; indels, Extended Data Fig. 3a–c). Simple rules were established to determine whether a call was validated, not validated or uninterpretable based on the existence of high-quality variant-supporting reads. We manually evaluated discordant calls and used additional evidence to resolve these discrepancies (see Methods for details). Combining the three validation experiments, the validation rates for SNVs and indels in the reference call set (HighConf and MedConf) were 99.93% (1,417/1,418) and 97.50% (78/80), respectively (Table 2).

PacBio long-read sequencing might cover the genomic regions where short reads cannot be mapped, especially in the

high GC/AT or low-complexity genomic regions (Extended Data Fig. 2). Although its higher sequencing error rate than short-read sequencing makes it inappropriate for somatic mutation discovery (Supplementary Table 2), we used PacBio long-read sequencing to confirm 99.3% SNVs and 98.5% indels in HighConf. In addition, for high-VAF calls in low mapping score and low sequence complexity regions, we demoted their confidence level when they were not supported by PacBio data ($P < 0.05$ for one-sided two-proportion z test when PacBio had no supporting read), resulting in the removal of 33 SNVs and 11 indels from the reference call set due to discordance from PacBio data.

Discovery and validation of germline variants. For 21 WGS replicates with gDNA from HCC1395BL cells, we used three aligners (BWA-MEM, Bowtie2 and NovoAlign) and four germline variant callers (FreeBayes⁴³, Real Time Genomics (RTG) (<https://www.realtimegenomics.com/>), DeepVariant⁴⁴ and HaplotypeCaller⁴⁵) to call SNVs and indels. We used a generalized linear mixed model

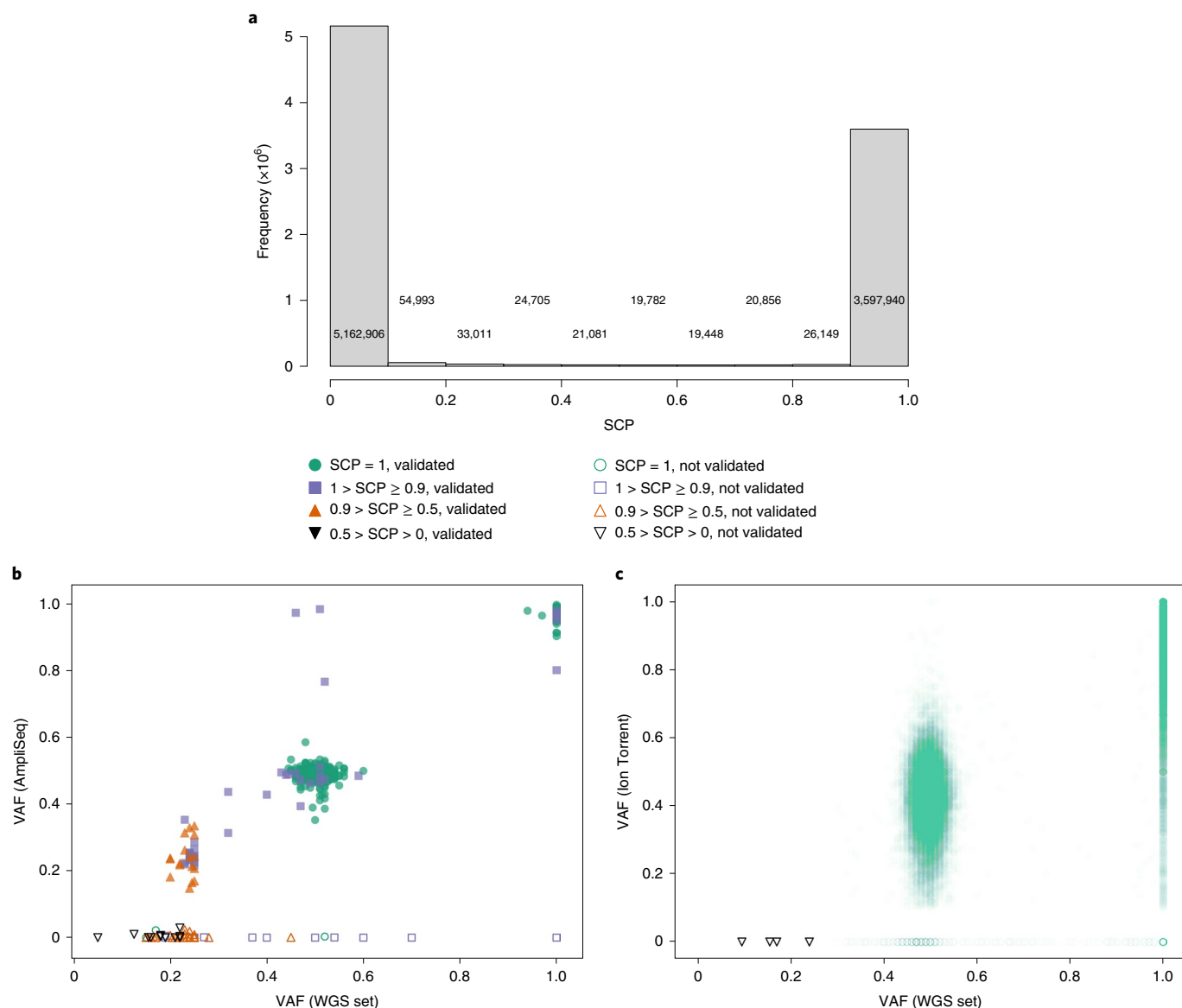


Fig. 3 | Initial definition and validation of germline variants. **a**, Histogram of SCP for germline variants identified by four callers from 63 BAM files. **b**, VAF scatter plot of germline SNVs by the call set and AmpliSeq data. Pearson's $R=0.986$ for SCP=1 call. **c**, VAF scatter plot of germline SNVs by the call set and Ion Torrent WES. Pearson's $R=0.758$ for SCP=1 call. In **b** and **c**, colors indicate the calling probability of the germlines variants, whereas shapes indicate their validation status.

(GLMM) to fit and consolidate various call sets (see the Methods for details). For SNVs and indels called at least four times, we estimated their SNV/indel calling probability (SCP) using the average over the four factors (sequencing center, sequencing replicate, aligner and caller). The frequency of SCPs showed a bimodal pattern (Fig. 3a). Most SNV calls (97%) had SCPs of <0.1 (57%) or >0.9 (40%). Only a few calls (3%) were between 0.1 and 0.9, indicating that when SNVs were called, 40% of them would be called as SNVs repeatedly, and the others would be called only infrequently.

We validated a subset of germline variants using AmpliSeq and Ion Torrent. For the highest confidence calls (SCP=1; that is, they were called in each combination of replicate, aligner and caller), the validation rates for SNVs and indels were 99% and 98% for AmpliSeq and 98% and 97% for Ion Torrent (Supplementary Table 3). Of the 11 SNVs with SCP values <0.5, none were confirmed by AmpliSeq.

Most validated germline variants had VAFs around 50% or 100% (Fig. 3b,c). By contrast, a considerable number of lower-confidence

germline SNVs were clustered around a VAF of 20% and could not be validated (Fig. 3b). VAF scatter plots for indels were qualitatively similar to those of the SNVs (Extended Data Fig. 4).

Heterogeneity of the HCC1395 cancer cell line. Previous studies of the TNBC cell line revealed large-scale genomic instability and ploidy changes²⁴, which were confirmed by our karyotype and cytogenetic analysis of the HCC1395 cell line (Extended Data Figs. 5 and 6). The WGS CNA analysis by ascatNgs⁴⁶ consistently revealed numerous chromosome gains and losses as well as LOH events (Extended Data Fig. 7). In addition, we observed peaks of VAF at 0.15 and 0.08, suggesting that these SNVs were from subclones (Extended Data Fig. 8).

We performed clonality analysis⁴⁷ leveraging WGS and WES data derived from bulk DNA as well as using single-cell data from 10x Genomics Chromium CNV analysis. The phylogenetic tree (Fig. 4a) generated by PhyloWGS⁴⁸ suggests a branching evolution that is

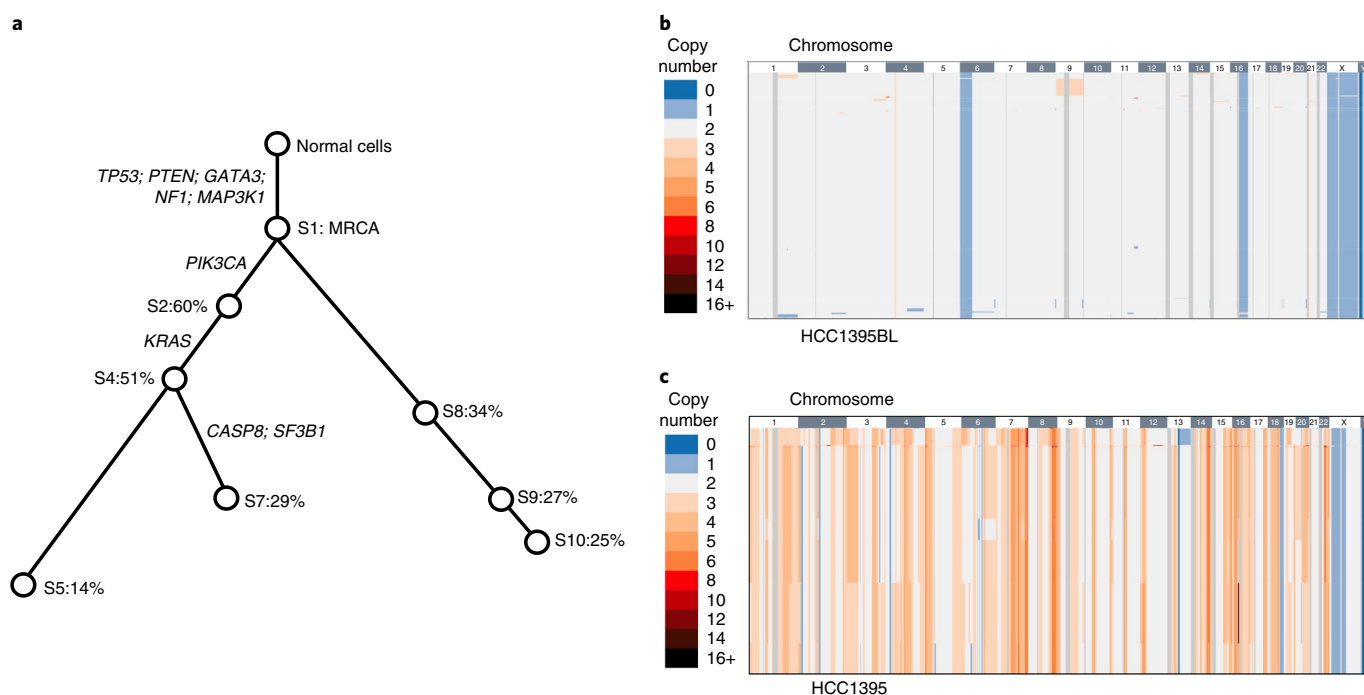


Fig. 4 | Clonality analysis of the HCC1395 cell line using bulk DNA and DNA from single cells. a, The inferred tumor phylogenetic tree. The subclone S1 represents the most recent common ancestor (MRCA) of all tumor cells, and S2 to S10 represent the subclones with various cancer cell fractions (for example, S2: 60%). The edges represent the evolutionary relationships between subclones. Subclones S3 and S6 are not shown given that their cancer cell fractions were less than 10%. Most point mutations are in driver genes (labeled beside the edges) present in the MRCA. Using the 10x Genomics Single Cell CNV Solution, integer-scaled CNA profiles were obtained across the genomes of 638 HCC1395BL cells (**b**) and 1,270 HCC1395 cells (**c**). Noisy cells and cells in the S phase of the cell cycle were removed. The complete linkage method was used for hierarchical clustering. Each row represents a cell being sequenced; similar cells were clustered together based on CNVs. Chromosome-scale gains are in orange and losses are in blue in the heat maps.

consistent with previous reports^{49–52}; the clonal expansion was further confirmed by superFreq⁴⁷ (Extended Data Fig. 9a). Specifically, HCC1395 cells showed some typical characteristics of breast cancer tumor development, including a clonal driver mutation in *TP53*, clonal copy neutral loss of heterozygosity (NLOH) at chromosome 17q and an early driver mutation in *PIK3CA*. We detected most of the point mutations in driver genes and almost all of the somatic CNAs involving driver genes present in the most recent common ancestor clone S1, suggesting that most of the driver events are clonal^{53,54}. In addition, the branching phylogenetic tree suggests that the tumor continues clonal evolution, resulting in two major branches (beginning with nodes at S2 and S8) either in vivo or in cell culture.

We further inferred both clonal and subclonal CNAs using subHMM⁵⁵ (Extended Data Fig. 9b). The estimated purity based on CNAs was 99%, as expected for a typical breast cancer cell line. The existence of subclonal CNAs was also observed in single-cell gDNA sequencing of 1,270 cells for HCC1395 and 638 cells for HCC1395BL (Fig. 4b,c). Integer-scaled CNA profiles across the genomes showed substantial subclonal CNAs in the TNBC cell line (Fig. 4c). Our CNA analyses revealed that HCC1395 is a heterogeneous cell line.

Discussion

Our reference samples and their call sets can have broad applications in benchmarking sequencing pipelines. For example, we used the reference call sets to benchmark WGS and WES in companion studies^{56,57}, which compared various experimental conditions to determine factors encompassing both analytical technology platforms and bioinformatic methods that affect the reproducibility and accuracy of somatic mutation detection. In particular, the variants in high-confidence regions can help to assess detection sensitivity and specificity. Our reference samples have also been used

to benchmark single-cell RNA-sequencing technologies⁵⁸, and the associated data are available to the community⁵⁹. The characteristics of genetic alterations in HCC1395 cells are in line with what have been described in a large cohort study⁶⁰ (Extended Data Fig. 10). The changes in ploidy observed in HCC1395 cells are common for breast cancers⁶¹. Therefore, the large number of somatic mutations and CNAs in HCC1395 cells provides the genomic representativeness desired in a reference gDNA sample. In addition, many variants and mutations in HCC1395 cells have clinical implications. In the coding regions, 207 somatic mutations in the reference call set are documented in the COSMIC database, and 8 germline variants are annotated as pathogenic in the ClinVar database. Interestingly, we found nonsense germline variants and somatic mutations in *BRCA1* and *BRCA2* genes, respectively, and hotspot somatic mutations in the *TP53* and *FGFR1* genes (Supplementary Data File 2). Hence, our reference call sets might also be valuable to benchmark bioinformatics pipelines for the detection of mutations with targeted panels.

Genetic materials derived from the paired tumor–normal cell lines are sustainable for community usage. We chose cell lines over real tissues to extract gDNA as somatic reference samples because real tissues are not renewable. To address the same consideration, the GIAB Consortium, which has been leading the efforts in establishing standard reference materials for germline variation detection, chose to use cell lines rather than real tissues from healthy individuals^{11,62}. In addition, the spatial heterogeneity of tumor tissue^{63,64}, for which different slices of the same tissue might have different mutation profiles, hampers the assessment of sequencing assay reproducibility. To mitigate genetic drift during the expansion of cell lines⁶⁵ and related concerns, GIAB generated a large quantity of gDNA from normal cell lines as reference material so that DNA aliquots could be distributed to the community. However, new

material will eventually need to be qualified when the current stock of DNA is exhausted.

An alternative approach adopted by this study to establish cell substrate as a source of reference materials is through a cell banking tier system, which has been fully documented⁶⁶. In this study, we made a large batch of gDNA from each of the two cell lines and distributed aliquots to the sequencing centers to develop reference call sets. The remaining gDNA can be used for validating or benchmarking any emerging sequencing technologies or protocols. Should the need for additional gDNA arise, cell aliquots derived from the same batch of the reserved master cells can be used for the production of gDNA, and genetic drift can be characterized with a bridging study using our gDNA materials and the reference call sets. We propose the call sets reported here as a reference and expect that the reference call sets and high-confidence regions might be further refined as sequencing technologies and data analysis tools improve.

There are several limitations with the somatic reference call sets reported here. First, reference somatic mutations were called in regions of the genome limited to those that could be analyzed with the Illumina sequencing platforms. Although we used Ion Torrent as an orthogonal method to validate some mutations, other mutations were validated on the Illumina sequencing platform. PacBio long-read sequencing with modest sequencing depths was also applied for validation; however, its sequencing error rates were higher than some low-VAF variants in the reference call sets (Supplementary Table 2). Second, the reference genome alignment-based approach might also leave undetected somatic mutations within regions that are susceptible to variations in a personal genome. Indeed, our ongoing study using a personal genome (that is, de novo assembly of the genome for the normal cell line) indicated that extra new mutations could be identified using a personal genome as the reference genome⁶⁷. Third, reproducibility was used as a proxy for defining a reference call set that can minimize biases/artifacts from library preparation, sequencing machine and bioinformatics pipeline. Our reference call set was defined with a tumor-normal match approach for gDNA derived from fresh cell cultures. To benchmark the performance of other sample types, such as FFPE or ctDNA, consideration needs to be taken to watch out for artifacts due to DNA damage (for FFPE)⁶⁸ or clonal hematopoiesis (for ctDNA)⁶⁸. Fourth, our somatic reference call set is not meant to be a comprehensive catalog of genomic alterations in breast cancer but, instead, is meant to serve as an analytical benchmark reference with many types of genomic alterations.

In summary, we established well-characterized somatic mutation and germline reference call sets for a pair of tumor-normal cell lines and shared them with the scientific community so that they can be used for evaluating NGS sequencing analysis pipelines. The diverse sequencing data generated from multiple platforms at multiple sequencing centers can help tool developers to build more accurate and robust artificial intelligence models for somatic mutation detection. The preserved gDNA and master cells can be used to develop standard reference materials in the future for assay development, qualification, validation and proficiency testing. Furthermore, the methodology used in this study can be extended to establish reference materials and reference datasets for additional sample types.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41587-021-00993-6>.

Received: 2 May 2019; Accepted: 18 June 2021;
Published online: 9 September 2021

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Methods

Cell lines and DNA extraction. HCC1395 human breast carcinoma cells (expanded from ATCC CRL-2324) were cultured in ATCC-formulated RPMI-1640 medium (ATCC, 30-2001) supplemented with fetal bovine serum (FBS; ATCC, 30-2020) to a final concentration of 10%. Cells were maintained at 37°C with 5% CO₂ and were subcultured every 2–3 d, per ATCC recommended procedures, using 0.25% (wt/vol) trypsin–0.53 mM EDTA solution (ATCC, 30-2101) until appropriate densities were reached. HCC1395BL human B lymphoblast Epstein–Barr virus-transformed cells (expanded from ATCC CRL-2325) were cultured in ATCC-formulated Iscove's Modified Dulbecco's medium (ATCC, 30-2005) supplemented with FBS to a final concentration of 20%. Cells were maintained at 37°C with 5% CO₂ and were subcultured every 2–3 d, per ATCC recommended procedures, using centrifugation with subsequent resuspension in fresh medium until appropriate densities were reached. Final cell suspensions were centrifuged and resuspended in PBS for nucleic acid extraction.

All cellular genomic material was extracted using a modified phenol–chloroform–isoamyl alcohol extraction approach. Cell pellets were resuspended in TE, lysed in a 2% Triton X-100/0.1% SDS/0.1 M NaCl/10 mM Tris/1 mM EDTA solution and extracted with a mixture of glass beads and phenol–chloroform–isoamyl alcohol. After multiple rounds of extraction, the aqueous layer was treated further with chloroform–isoamyl alcohol and underwent RNase treatment and DNA precipitation using sodium acetate (3 M, pH 5.2) and ice-cold ethanol. The final DNA preparation was resuspended in TE, and aliquots were distributed to sequencing centers for NGS and microarray analysis.

Cell line karyotyping. Karyotyping was performed by Cell Line Genetics as described previously⁶⁹. Cells were treated with Colcemid (Gibco) for 40 min, exposed to 0.075 M KCl for 23 min at 37°C and then fixed with 3:1 methanol:glacial acetic acid. Slides were stained with Leishman's stain before observation. During observation, roughly 20 metaphase cells were counted on the microscope, and numerical and structural chromosome aberrations were recorded. An analysis of 5–10 band cells was performed at the microscope using a ×100 objective with an effort to karyotype at least two cells from each clone.

WGS and WES on Illumina platforms. See methods section in ref. ⁵⁶.

PacBio library preparation and sequencing. Fifteen micrograms of material was sheared to 40 kilobases (kb) with Megarupter (Diagenode). Per the Megarupter protocol, the samples were diluted to below 50 ng μl⁻¹. A 1× AMPure XP bead cleanup was performed. Samples were prepared as outlined in the PacBio protocol titled 'Preparing >30 kbp SMRTbell Libraries Using Megarupter Shearing and Blue Pippin Size-Selection for PacBio RS II and Sequel Systems'. After library preparation, the library was run overnight for size selection using the Blue Pippin (Sage). The Blue Pippin was set to select a size range of 15–50 kb. After collection of the desired fraction, a 1× AMPure XP bead cleanup was performed. The samples were loaded on the PacBio Sequel (Pacific Biosciences) following the protocol titled 'Protocol for loading the Sequel'. The recipe for loading the instrument was generated by the PacBio SMRTLink software v5.0.0. Libraries were prepared using Sequel chemistry kits v2.1, a SMRTbell template kit 1.0 SPv3, a magbead v2 kit for magbead loading, sequencing primer v3 and SMRTbell cleanup columns v2. Libraries were loaded between 16:00 and 20:00.

AmpliSeq deep sequencing. Criteria for picking variants. The variant call format (VCF) files were filtered to generate a list of variant coordinates that could be uploaded into Illumina Design Studio. The list of somatic variants was selected based on the following criteria: all variant call confidence tiers were represented equally, low and high VAFs were covered and all chromosomes were represented. The logic behind picking the variants was to have an equal number of variants per tier while equally representing all four confidence levels. The selection process also attempted to ensure representation of all chromosomes across all tiers but did not require that all chromosomes be represented in each tier. The TNBC cell line HCC1395 has significant structural rearrangements and ploidy changes, resulting in severe overrepresentation of variants from chromosomes 6, 16 and X. Therefore, variants from these three chromosomes were downweighted in random sampling to ensure that variants from other chromosomes had a statistical chance of being selected. Germline variants were also included to determine the accuracy of the AmpliSeq panel with reference to previously characterized variants in cell line HCC1395. At the end of this process, a total of 2,477 variants (somatic and germline SNVs and indels) were selected to enter into the AmpliSeq design software.

The 2,477 variants' chromosomal positions were expanded to 300-bp regions to upload into Illumina Design Studio, selecting for a 275-bp amplicon size. The design algorithm failed to generate complete coverage for 1,109 of the 300-bp regions, leaving 1,368 regions with the following makeup: 641 germline variants, 77 germline indels, 554 somatic variants and 96 somatic indels. The 1,368 300-bp amplicon regions were submitted to Illumina for panel manufacturing.

Library construction and sequencing. The HCC1395 and HCC1395BL libraries were prepared in triplicate as specified in the Illumina protocol (document number 1000000036408 v04) following the two oligo pools workflow with

10 ng of input gDNA per pool. The numbers of amplicons per pool were 1,517 and 1,506, respectively. The libraries were quality checked using an Agilent TapeStation 4200 with the DNA HS 1000 kit and quantitated using a Qubit 3.0 and DNA high-sensitivity assay kit. The libraries were applied to a MiSeq v2.0 flowcell. They were then amplified and sequenced with a MiSeq 300 cycle reagent cartridge with a read length of 2 × 150 bp. The MiSeq run produced 7.3 gigabases (gb) (94.5%) at ≥Q30. The number of reads passing filtering was 48,886,352 of 52,601,972 total (96.4%).

Sequence alignment and variant validation analysis for AmpliSeq. We aligned the FASTQ files with BWA-MEM and counted the number of variant-supporting reads and total reads for each variant position with MQ ≥ 40 and BQ ≥ 30 cutoffs. The total sequencing depths for the tumor and normal samples were approximately 2,000×. The following rule was applied to the AmpliSeq data to determine whether a call was deemed confirmed, not confirmed or uninterpretable:

- If tumor and normal depth were both ≥600× and if variant depth in the tumor was ≥100 and normal variant depth was <10 → validated.
- Else, if normal depth was ≥600× and if variant reads in normal consisted of ≥10% of the total depth → not validated (germline false positive).
- Else, if tumor total depth was ≥1,000 and tumor variant depth was <5, or if tumor total depth was ≥1,000 and tumor variant depth consisted of <0.1% of the total depth → not validated (no variant reads in tumor when there should be).
- Else, if tumor or normal total depth was ≤50 → uninterpretable.
- Else → manual inspection on Integrative Genome Viewer (IGV) to make a determination.

In addition, we used IGV to study calls that were

1. Originally annotated as 'confirmed' but had a large difference in VAF calculated from the truth set data and AmpliSeq data and were reannotated if needed.
2. HighConf calls that were 'not confirmed', although no annotation was actually changed.
3. Unclassified calls that were 'confirmed' and reannotated if needed.

When there was manual reannotation after IGV inspection, a comment was left in the final column in the validation file.

Ion Torrent WES. Library construction and sequencing. A SureSelect Target Enrichment Reagent kit, PTN (G9605A), SureSelect Human All Exon v6 + UTRs (5190-8881), Hercules II Fusion DNA Polymerase (600677) from Agilent Technologies and an Ion Xpress Plus Fragment kit (4471269, Thermo Fisher Scientific) were combined to prepare libraries according to the manufacturer's guidelines (user guide: SureSelect Target Enrichment System for Sequencing on Ion Proton, version C0, December 2016, Agilent Technologies). Before, during and after library preparation, the quality and quantity of gDNA and/or libraries were evaluated using a Qubit fluorometer 2.0 with a dsDNA HS Assay kit (Thermo Fisher Scientific) and an Agilent Bioanalyzer 2100 with a high sensitivity DNA kit (Agilent Technologies). For sequencing the WES libraries, the Ion S5 XL Sequencing platform with Ion 540-Chef kit (A30011, Thermo Fisher Scientific) and the Ion 540 Chip kit (A27766, Thermo Fisher Scientific) were used. One sample per 540 Chip kit was sequenced, generating up to 60 million reads with an average length of 200 bp.

Sequence alignment and analysis of Ion Torrent data. Raw reads were first filtered for low-quality reads and trimmed to remove adapter sequences and low-quality bases. This step was performed using the BaseCaller module of the Torrent Suite software package v5.8.0 (Thermo Fisher Scientific). Low-quality reads were retained from further analysis in the raw signal processing stage. Low-quality bases were trimmed from the 5' end if the average quality score of the 16-base window fell below 16 (Phred scale), cleaving 8 bases at once. Processed reads were mapped to the GRCh38 reference genome by the TMAP module of the Torrent Suite software package using the default map4 algorithm with recommended settings.

Variant validation analysis. The following rule was applied to interpret WES validation by both Ion Torrent and HiSeq with the same MQ ≥ 40 and BQ ≥ 30 thresholds as used for AmpliSeq data.

- If tumor variant depth was >2 and tumor VAF was >10 times the normal VAF → confirmed.
- If normal total depth was >20 and normal VAF was >10% → not confirmed (germline).
- If the expected value for variants in reads in the tumor is >5 based on truth set VAF and tumor total depth but there was no such read → not confirmed (no signal when signal was expected).
- Else → uninterpretable.

Bioinformatics pipelines. In this section, we describe the bioinformatics analysis methods to generate the high-confidence somatic mutation call set. The overall schematic is shown in Fig. 1. The commands and source codes are documented at our project site at <http://sites.google.com/view/seqc2>.

Read alignment. For each of the paired-end read files (that is, FASTQ 1 and 2 files) generated by Illumina sequencers (HiSeq, NovaSeq and MiSeq platforms), we first trimmed low-quality bases and adapter sequences using Trimmomatic²⁰ (performed at Frederick National Laboratory for Cancer Research). The resulting sequences were then mapped with three different aligners (that is, BWA-MEM, Bowtie2 and NovoAlign) to create three BAM files. Picard Tools (<http://broadinstitute.github.io/picard/>) was then used to mark PCR and optical duplicates in the BAM files (except for AmpliSeq data from MiSeq).

The Ion Torrent reads were first trimmed for low-quality bases and adapter sequences using the BaseCaller module of the Torrent Suite software package v5.8.0. Low-quality bases were trimmed from the 5' end if the average quality score of the 16-base window fell below a BQ of 16, cleaving 8 bases at once. The processed reads were then mapped with TMAP using the default map4 algorithm with otherwise default settings. Picard Tools was then used to mark PCR and optical duplicates on the BAM files.

Building the center- and aligner-specific SomaticSeq classifiers. BAM files produced from the same sequencing centers and platform and aligned with the same aligner were grouped into one data group. For instance, the three pairs of HiSeq BAM files at IL aligned with BWA-MEM were grouped into one data group, and the same reads aligned with NovoAlign were grouped into another data group under NovoAlign (Fig. 1). BAM files that came from LL, NC and EA were grouped into 'Others'. Hence, there were five data groups for each aligner: (1) HiSeq at IL, (2) NovaSeq at IL, (3) HiSeq at FD, (4) HiSeq at NV and (5) HiSeq at Others. SNV and indel SomaticSeq classifiers were built for each of the first four data groups because we took advantage of multiple sequencing replicates of the normal genome to build machine learning classifiers that remove data group-specific artifacts as false positives. Because three aligners were used, 12 SNVs and 12 indel center- and aligner-specific classifiers were created. For the Others groups, data from the first four groups were combined to simulate aligner-specific (but not center-specific) artifacts for it.

For each data group that has intracenter sequencing replicates, we designated normal replicate 1 as the normal and normal replicate 2 as the pseudotumor. BAMSurgeon was used to spike approximately 100,000 in silico SNVs and 20,000 in silico indels into the pseudotumor to create an in silico tumor¹⁸. SomaticSeq's tumor-normal workflow was then used for the in silico tumor-normal pairs (Supplementary Fig. 1a). Briefly, six somatic mutation callers were incorporated into the workflow: MuTect2 (ref. ¹³), SomaticSniper³⁶, VarDict³⁷, MuSE³⁸, Strelka2 (ref. ³⁹) and TNscope⁴⁰. About 100 genomic and sequencing features were extracted for each variant call using SomaticSeq¹². Every mutation call that was not spiked in by BAMSurgeon was labeled a false positive, and only the in silico variants were labeled true positives.

Two training sets were created for each data group. Training set A designated normal replicate 2 as the in silico tumor versus normal replicate 1. Training set B designated normal replicate 3 as the in silico tumor versus normal replicate 2. Different mutations were spiked into the two different training sets. We have done cross-validation between these two sets, that is, classifiers built on training set A were used to classify in silico tumor-normal pairs in training set B and vice versa. Cross-validations were to ensure that SomaticSeq classifiers could filter out false positives in different datasets from the same data group reliably and not just the same dataset it was trained on.

Supplementary Fig. 1b displays the classification accuracy of cross-validation from the $12 \times 2 = 24$ SNV cross-validations. PASS calls had SomaticSeq SCORE values ≥ 0.7 , and, conversely, REJECT calls had SCORE values ≤ 0.1 . Calls with $0.1 < \text{SCORE} < 0.7$ were labeled LowQual. In the 24 cross-validations, the classification sensitivity (that is, true positives classified as PASS in the combined call set from the six callers), specificity, positive predictive value and negative predictive value were 98.39%, 99.23%, 99.52% and 99.86% for SNVs and 96.96%, 98.03%, 98.08% and 99.67% for indels, respectively. A few in silico mutations were not detected by any of the six callers. They are discussed in the Supplementary Materials. For each data group, training sets A and B were combined to make one SNV and one indel SomaticSeq classifier for later use. For the Others groups, training sets from the other four data groups were combined to make an SNV and an indel classifier.

Building aligner-specific NeuSomatic classifiers. For NeuSomatic, instead of creating center-specific and variant type-specific (that is, SNV and indel separately) classifiers, one NeuSomatic-Ensemble and one NeuSomatic-Standalone classifier were created for each aligner by combining all the simulated data from all the data groups.

Make somatic mutation calls in each HCC1395 versus HCC1395BL pair. The same six somatic mutation callers were run on each pair of tumor-normal BAM files performed on the Cancer Genomics Cloud (CGC) by Seven Bridges Genomics. The corresponding SomaticSeq and NeuSomatic classifiers (described previously) were used to classify mutations for each of the 63 call sets, creating somatic SNV and indel VCF files for each tumor-normal pair (Supplementary Fig. 3a). The histograms of the variant scores (Supplementary Fig. 3b,c) for the real datasets

are qualitatively similar to the cross-validation of the training datasets shown in Supplementary Fig. 1b,c.

Determine confidence level for initial somatic mutation call with SomaticSeq. The schematic of this process is shown in Fig. 1. The four confidence levels (that is, HighConf, MedConf, LowConf and Unclassified) of the somatic mutation calls were annotated primarily based on the three aligner-centric classifications of a somatic mutation call. The aligner-centric classification of a call for each aligner was in turn determined by the five data group scores within this aligner. Each data group score was in turn determined by a call's reproducibility across the sequencing replicates within the data group.

The algorithm is described in detail in the Supplemental Material. For example, if a call was classified as a PASS in two replicates and LowQual in the other replicate for HiSeq at IL with BWA-MEM (a data group), it would have had a data group score of +3 (Supplementary Fig. 7). If the five data group scores added up to at least 6, it would have been considered 'Strong Evidence' for BWA-centric classification (Supplementary Fig. 7). If this call was deemed 'Strong Evidence' for BWA-, Bowtie2- and NovoAlign-centric classifications, it would have been considered a HighConf call (Supplementary Fig. 7). For low-VAF ($\leq 15\%$) calls initially annotated as LowConf and Unclassified, additional 350x and 400x (combined nine NovaSeq at IL replicates) datasets were used to potentially rescue those calls into MedConf if they were deemed PASS in those high-depth datasets. The 15% VAF threshold for confidence-level recalibration was determined after examining AmpliSeq deep sequencing data when it was determined that for non-HighConf calls with $\text{VAF} \leq 15\%$, they were far more likely to be true positives than false positives. HighConf and MedConf calls were grouped into the truth set. Calls in the truth set had many samples classified as PASS across different aligners and sequencing centers with very few (if any) REJECTS.

Determine high-confidence genomic regions and somatic mutation reference call set. We have labeled four different confidence levels in our entire call set (the super set): HighConf, MedConf, LowConf and Unclassified. HighConf and MedConf calls were considered as the reference call set used to benchmark pipelines along with the high-confidence genomic regions. Twenty-base pair regions spanning LowConf and Unclassified calls were not included in the high-confidence region to ensure that there were no false negatives in the high-confidence regions. The technical details of the workflow are described in detail in the Supplemental Material. The commands and future updates will be documented on the Working Group's project website at <https://sites.google.com/view/seqc2>. In this section, we will describe general methods to categorize the somatic mutation in the following seven steps.

1. Use corresponding SomaticSeq classifiers to score somatic mutation calls in each of the 63 tumor-normal data sets. Combine the call sets and keep all mutation calls that have been scored ≥ 0.7 in at least 1 of the 63 datasets.
2. Assign initial confidence levels based on how consistent a mutation call has been classified by SomaticSeq as PASS across the three aligners and six sequencing centers as well as false-positive markers, such as MQ, read-to-reference mismatches, BQ and inconsistent VAF movement with the 350x tumor-normal titration series.
3. Promote the initial confidence levels and rescue additional low-VAF calls using 350x and 400x SomaticSeq datasets.
4. Modify confidence levels of calls when SomaticSeq and NeuSomatic classifications showed significant discrepancies or when the SomaticSeq classifications themselves were not consistent across the 63 datasets. Rescue additional calls that were captured by NeuSomatic but not SomaticSeq.
5. Use the 1,500x NeuSomatic tumor-normal call sets (combining all short-read WGS reads) to promote low-VAF calls and rescue low-VAF calls that were not captured previously.
6. For HighConf and MedConf calls that are not supported by PacBio data ($P < 0.05$, two-proportion z test), demote them to LowConf if the variant in question is in a low mapping or low sequence complexity region.
7. Finally, label HighConf and MedConf calls in the consensus-callable regions as PASS. Create high-confidence regions by removing 20-bp regions spanning LowConf and Unclassified positions and the germline LOH of the three chromosome arms (chromosome 6p, chromosome 16q and chromosome X) from the consensus-callable regions.

Validate somatic mutations with tumor titration series fitting scores. We divided SNVs into $k = 13$ groups based on the VAF, $M_k = 0.00, 0.05, 0.10, \dots, 0.55$ and ≥ 0.60 at 100% purity. For the k th group, there are n_k SNVs, each of which has seven VAF y_{ij}^{observed} , $i = 1, 2, \dots, 7$ diluted purities 100%, 75%, 50%, 20%, 10%, 5% and 0%. For SNVs in the k th group, we fitted a linear mixed model because each SNV has its VAF calculated at diluted purities and obtained the best unbiased predictor y_{ij}^{fitted} . We calculated the relative squared error

$$\sqrt{\sum_{j=1}^7 \left(y_{ij}^{\text{observed}} - y_{ij}^{\text{fitted}} \right)^2 / M_k}$$

The mean relative squared error (MRSE) is simply the average of relative squared error for SNVs in the k th group,

$$\text{MRSE}_k = \frac{1}{k} \sum_{j=1}^{n_k} \sqrt{\sum_{i=1}^{j=7} (Y_{ij}^{\text{observed}} - Y_{ij}^{\text{fitted}})^2 / M_k} \quad (1)$$

Tumor purity fitting score is the inverse of MRSE. MRSE was applied to every somatic SNV call in the super set. The 'error bars' were calculated by removing data points whose $Y_{ij}^{\text{observed}} - Y_{ij}^{\text{fitted}}$ values were 1 s.d. below the mean (upper bar) or 1 s.d. above the mean (lower bar).

Determine callable regions. The consensus high-confidence region was determined in the following manner:

1. GATK CallableLoci was run on each tumor and normal ($63 \times 2 = 126$) BAM file.
2. Within each data group:
 - a. Determine callable regions in the majority of the tumor replicates.
 - b. Determine callable regions in the majority of the normal replicates.
 - c. Consider the callable regions of the data group to be the intersection of the regions in (a) and (b).
3. Within each of the three aligners, consider regions deemed callable by the majority (that is, three of five) of the data groups to be the callable regions of the aligner.
4. Consider regions deemed callable by the majority (that is, two of three) of the aligners to be the consensus-callable regions. Label somatic mutation calls outside the consensus regions 'NonCallable' as a value in FLAGS in the INFO field.

Determine germline variants. We used four germline variant callers (freebayes⁴³, RTG, DeepVariant⁴⁴ and HaploTypeCaller⁴⁵), which were readily available on the National Institutes of Health (NIH) Biowulf cluster. We ran each of them using the default parameters or parameters recommended by the user's manual. Specifically, for HaploTypeCaller, we included a flag for 'stand_call_conf 30' and used dbsnp build 146 on the GRCh38 reference genome. For DeepVariant, we used model.cpkt for WGS, which was provided by DeepVariant (gs://deepvariant/models/DeepVariant/0.5.1/DeepVariant-inception_v3-0.5.1+data-wgs_standard/).

To consolidate all the call sets, we fit a GLMM for each SNV that was sequenced at different sites on various replicates, aligned by three aligners and called by four callers. The SNVs considered were called at least four times by various combinations of factors (including sites, replicates, aligners and callers). We defined the SNV call probability P_{mnjk} as the probability of an SNV being called by a given aligner (α_{1m}) and a caller (α_{2n}) at a site (α_{3j}) and on a replicate within a site (α_{4k}). We modeled P_{mnjk} as

$$\log \frac{P_{mnjk}}{1 - P_{mnjk}} = \beta + \alpha_{1m} + \alpha_{2n} + \alpha_{3j} + \alpha_{4k} \quad (2)$$

Here, β is a fixed effect, and α_{1m} , α_{2n} , α_{3j} and α_{4k} are normally distributed random effects with zero means and variances α_1^2 , α_2^2 , α_3^2 and α_4^2 , respectively. The SNV call probability averaged across factors is defined as $P = \frac{e^\beta}{1 + e^\beta}$, and variance explained by each factor is defined as $\frac{\alpha_i^2}{\alpha_1^2 + \alpha_2^2 + \alpha_3^2 + \alpha_4^2}$, where $i = 1, 2, 3$ and 4. All these parameters were estimated by the R package lme4.

SNV annotation and TMB benchmarking. SnpSift was used to link dbSNP build 146 and COSMIC v85 identifiers, and information to the somatic variant calls and ClinVar (version 20190305) to the germline variant calls⁸. SnpEff was used to annotate genes and their functional effects (that is, missense, synonymous and on) of the variants⁷¹.

When counting non-synonymous variants, only missense_variant (319), stop_gained (16), stop_lost (1) and structural_interaction_variant (6) were counted. We used University of California Santa Cruz (UCSC) genome track GRCh38-based GENCODE v29 coding exons to obtain the coding regions (<http://genome.ucsc.edu/cgi-bin/hgTables?command=start>).

When hg19 coordinates were given by oncology panel vendors for target regions, we used <https://www.ncbi.nlm.nih.gov/genome/tools/remap> to remap them into hg38. The region size of the panels (as well as UCSC coding regions) had the three LOH arms (chromosome 6, chromosome 16 and chromosome X arm losses in normal) subtracted, and the mutation rates were calculated based on those region sizes.

CNV analysis with WGS. We used the CNA detection tool ascatNgs (version 4.2.1), which contains the Cancer Genome Project's Singularity workflow implementation of the ascatNgs⁴⁶. The tool was available on the NIH high-performance computing resource (Biowulf). The BAM files were generated

using BWA-MEM, and CNA was performed using the default parameters of ascatNgs recommended by the user's manual with a flag '-protocol WGS' specific for WGS data. The run command is 'ascat.pl -tumour tumor.bam -norm normal.bam -reference hg38.fa -snp_gc Snp_panel -protocol WGS'; the ASCAT SNP_panel was generated using a provided ascatSnpPanelGenerator Perl script with human genome reference hg38.

Cell line clonality analysis. To determine the clonality of HCC1395 and HCC1395BL cells, we performed somatic SNV and CNA analysis using superFreq⁴⁷ on capture WES datasets. Mapped and markDuplicate BAM files of a pair of HCC1395 and HCC1395BL datasets were used as input, and BAM files of the remaining replicates of the HCC1395BL library were used to filter background. An analysis was run using the superFreq default parameters. The clonality of each somatic SNV was calculated based on the VAF, accounting for local copy number. The SNVs and CNAs undergo hierarchical clustering based on the clonality and uncertainty across replicates for the tumor sample.

We inferred both clonal and subclonal CNA profiles using subHMM. First, we calculated log odds ratio (log OR) and log ratio (log R) values of variant allele counts for tumor versus normal. Second, log OR and log R were adjusted by library size and GC normalization. Third, subHMM embedded adjusted log OR and log R values into a hidden Markov model (HMM) and simultaneously conducted the segmentation and genotype mixture modeling to identify both clone/subclone region and region-specific clone/subclone genotype and clonal proportion in a heterogeneous tumor sample. Note that subHMM allows the subclonal genotype and proportion to vary across different subclonal regions. Hence, multiple subclonal regions with distinct clonal proportions are considered as distinct subclones.

Single-cell CNV and heterogeneity analysis. HCC1395 and HCC1395BL cells were cultured as described above. Cells (500,000) of each culture were suspended in 1 ml of suspension medium (10% DMSO in cell culture medium). Cells were collected the next day for single-cell CNV analysis via the 10x Genomics Chromium Single Cell CNV Solution (CG000153), which produces single-cell DNA libraries ready for Illumina sequencing, according to manufacturer's recommendations (10x Genomics). Libraries were sequenced on an Illumina HiSeq 4000 using standard manufacturer's protocols (Illumina) for 2×150 -bp paired-end reads. Demultiplexing BCL files from the sequencing run and CNV analysis was performed using 10x Genomics Cell Ranger DNA version 1.1 software. CNV and heterogeneity visualization analysis was performed via 10x Genomics Loupe scDNA browser.

Affymetrix Cytoscan HD microarray. We obtained DNA of two reference cell lines from ATCC (HCC1395, SCCL2324_D; HCC1395BL, SCCL2325_D). DNA concentration was measured spectrophotometrically using a Nanodrop (Life technology), and integrity was evaluated with a TapeStation 4200 (Agilent). Two hundred and fifty nanograms of gDNA was used to proceed with the Affymetrix CytoScan Assay kit (Affymetrix). The workflow consisted of restriction enzyme digestion with NspI, ligation, PCR, purification, fragmentation and end labeling. DNA was then hybridized for 16 h at 50 °C on a CytoScan array (Affymetrix), washed and stained in the Affymetrix Fluidics Station 450 (Affymetrix) and then scanned with the Affymetrix GeneChip Scanner 3000 G7 (Affymetrix). Data were processed with ChAS software (version 3.3). Array-specific annotation (NetAffx annotation release 36, built with human hg38 annotation) was used in the analysis workflow module of ChAS. Karyoview plot and segments data were generated with default parameters.

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Reporting Summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

All raw data (FASTQ files) are available on NCBI's SRA database (SRP162370). The call set for somatic mutations in HCC1395 cells, VCF files derived from individual WES and WGS runs, BAM files for BWA-MEM alignments and source codes are available on NCBI's ftp site (ftp://ftp-trace.ncbi.nlm.nih.gov/ReferenceSamples/seqc/Somatic_Mutation_WG). Some alignment files (BAM) are also available on Seven Bridges' CGC platform under SEQC-II project. gDNA tested in the current study was prepared by ATCC using cell expansions from master banks of cells for the HCC1395 (ATCC, CRL-2324) and HCC1395BL (ATCC, CRL-2325) cell lines. gDNA aliquots from these preparations were distributed to the sequencing centers to perform WGS and WES as described. For remaining gDNA aliquots, contact the corresponding authors. Contact ATCC for additional materials related to the HCC1395 and HCC1395BL cell lines.

Software and code availability

The code to create somatic reference call set v1.2 is deposited on GitHub under a BSD 2-Clause open-source license tagged at https://github.com/bioinform/somaticseq/tree/seqc2_v1.2. A snapshot can also be downloaded at https://github.com/bioinform/somaticseq/archive/seqc2_v1.2.tar.gz.

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Acknowledgements

We thank J. Zook of the National Institute of Standards and Technology for advice in establishing reference samples and truth sets, S. Gowrisankar of Novartis and S. Chacko of the Center for Information Technology, NIH, for their assistance with data transfer and J. Ye of Sentieon for providing the Sentieon software package. We also thank D. Goldstein of the Office of Technology and Science at the National Cancer Institute (NCI), NIH, and L. Amundadottir of the Division of Cancer Epidemiology and Genetics, NCI, NIH, for the sponsorship and the use of the NIH Biowulf cluster, R. Phillip, Y. Hu, S. Liang and Y. Li of the Center for Devices and Radiological Health, US FDA, for their advice on study design and manuscript writing, J. Collins and E. Stahlberg of Biomedical Informatics and Data Science Directorate at Frederick National Laboratory for Cancer Research for reviewing the manuscript and providing suggestions and Seven Bridges Genomics for providing storage and computational support on the CGC. B.Z. was supported by the Intramural Research Program of the NIH, NCI, Division of Cancer Epidemiology and Genetics. Y. Zhao, K.T., T.S., B.T., J.S. and Y.K. were supported by the Frederick National Laboratory for Cancer Research and through the NIH fund (contract number 75N910D00024). L. Shi and Y. Zheng were supported by the National Natural Science Foundation of China (31720103909), the National Key R&D Project of China (2018YFE0201600) and Shanghai Municipal Science and Technology Major Project (2017SHZDZX01). E.R. was supported by the European Union through the European Regional Development Fund (2014-2020.4.01.15-0012). The CGC has been funded in whole, or in part, by Federal funds from the NCI, NIH (HHSN261201400008C), and ID/IQ Agreement number 17X146 under contract number HHSN261201500003I. C.X. and S. Sherry were supported by the Intramural Research Program of the National Library of Medicine, NIH. This work also used the computational resources of the NIH Biowulf cluster (<http://hpc.nih.gov>). Original data were also backed up on the servers provided

by Center for Biomedical Informatics and Information Technology (CBIIT), NCI. The genomic work performed at the Loma Linda University (LLU) Center for Genomics was funded in part by the NIH grant S10OD019960, the American Heart Association grant 18IPA34170301, the Ardmore Institute of Health grant 2150141 and C.A. Sims' gift to LLU Center for Genomics. We acknowledge TopEdit for linguistic editing and proofreading during the preparation of this manuscript.

Author contributions

The study was conceived and designed by W.X., C.W., L. Shi, H.H., E.D., Z.T., B.N., W.T. and R.J. Biosample preparation was performed by L.K., K.L., M. Mars and T.H. NGS library preparation and sequencing was performed by W.C., Z.C., S. Stanbouly, K.I., H.J., E.J., G.P.S., Y. Zheng, B.T., Y.Y., J.S., Y.K., M. Mehat, V.P., M.S., T.H., E.P., R.K., J.D., P.V., R.M., D.G., S.K., E.R., A.S., J.N., U.L., J.W., J.L., P.D.H., C.C., S.M., J.S., J.F., D.B. and C.E.M. Data analysis was performed by L.T.F., W.X., B.Z., Y. Zhao, Z.Y., L.R., C.L., O.D.A., L. Song, J.L., T.S., K.T., D.M., C.N., M.C., S.M.E.S., M. Mohiyuddin, Y.G., L.Y., H.L., M.P., Z.L., W.S.L., J.K., J.A., E.T., V.Z., T.M. and J.T. Data management was performed by W.X., C.X. and S.T.S. The manuscript was written by L.T.F., B.Z., W.X., R.K., M. Moos, C.X., S.T.S. and Y. Zhao. W.X. managed the project.

Competing interests

L.T.F., S.M.E.S., M. Mohiyuddin, Y.G., L.Y. and H.L. are employees of Roche Sequencing Solutions Inc. L.K., K.L. and M. Mars are employees of ATCC, which provides cell lines and derivative materials. E.J., G.P.S. and O.D.A. are employees of Illumina Inc. V.P. and M.S. are employees of Novartis Institutes for Biomedical Research. T.H., E.P. and R. Kalamegham are employees of Genentech (a member of the Roche group). Z.L. is an employee of Sentieon Inc. R. Kusko is an employee of Immuneering Corp. C.C., S.M. and J.S. are employees of 10x Genomics. All other authors declare no competing interests.

Additional information

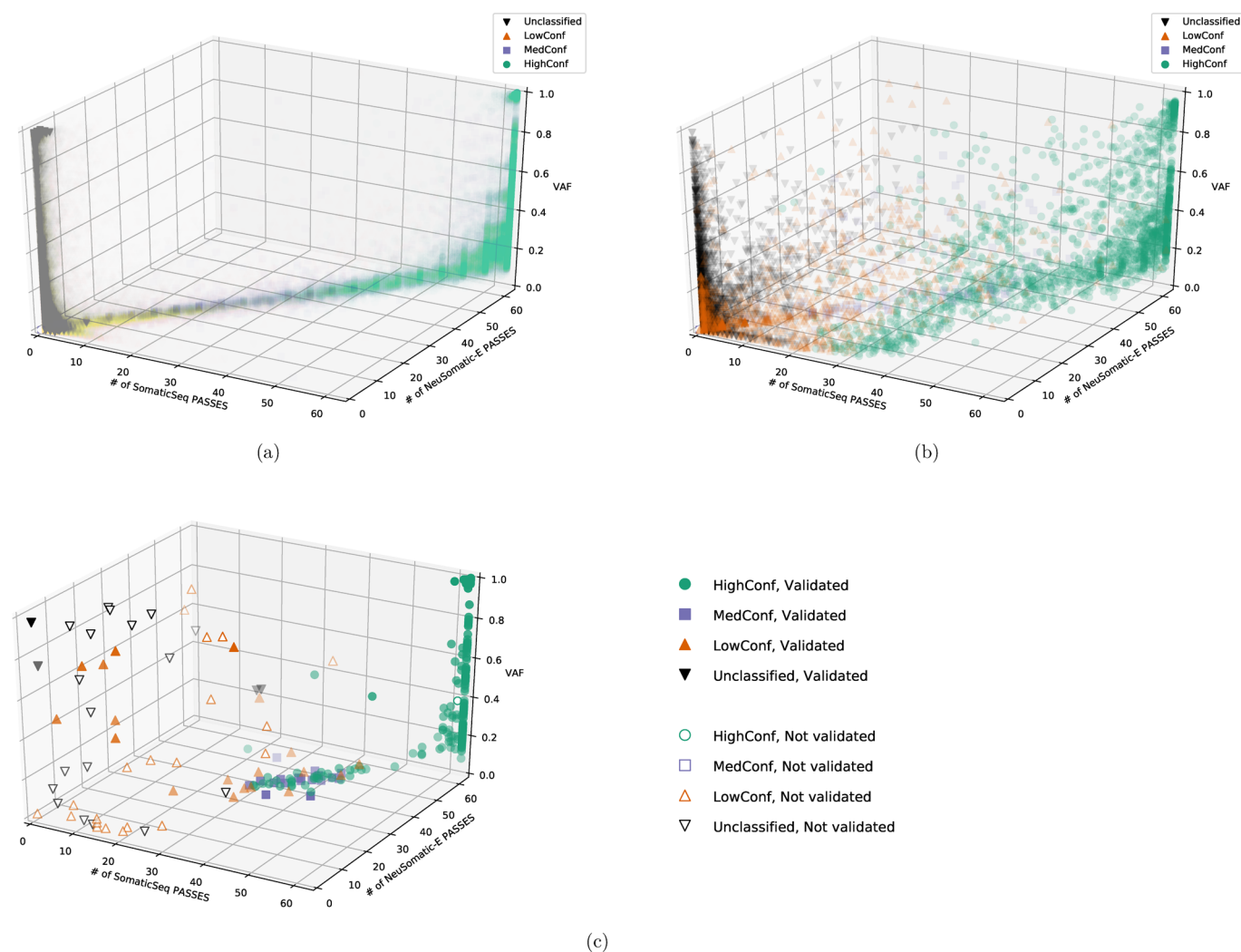
Extended data is available for this paper at <https://doi.org/10.1038/s41587-021-00993-6>.

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41587-021-00993-6>.

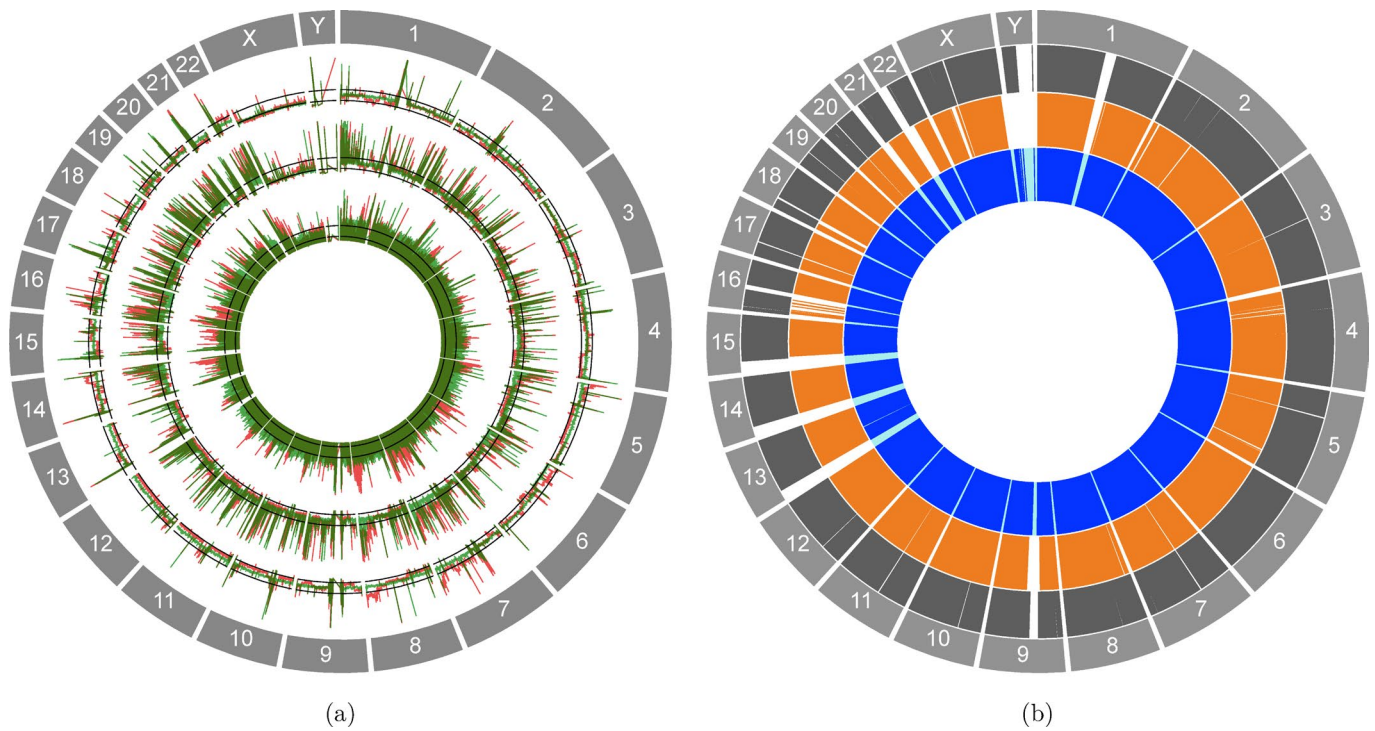
Correspondence and requests for materials should be addressed to H.H., L.S., C.W. or W.X.

Peer review information *Nature Biotechnology* thanks the anonymous reviewers for their contribution to the peer review of this work.

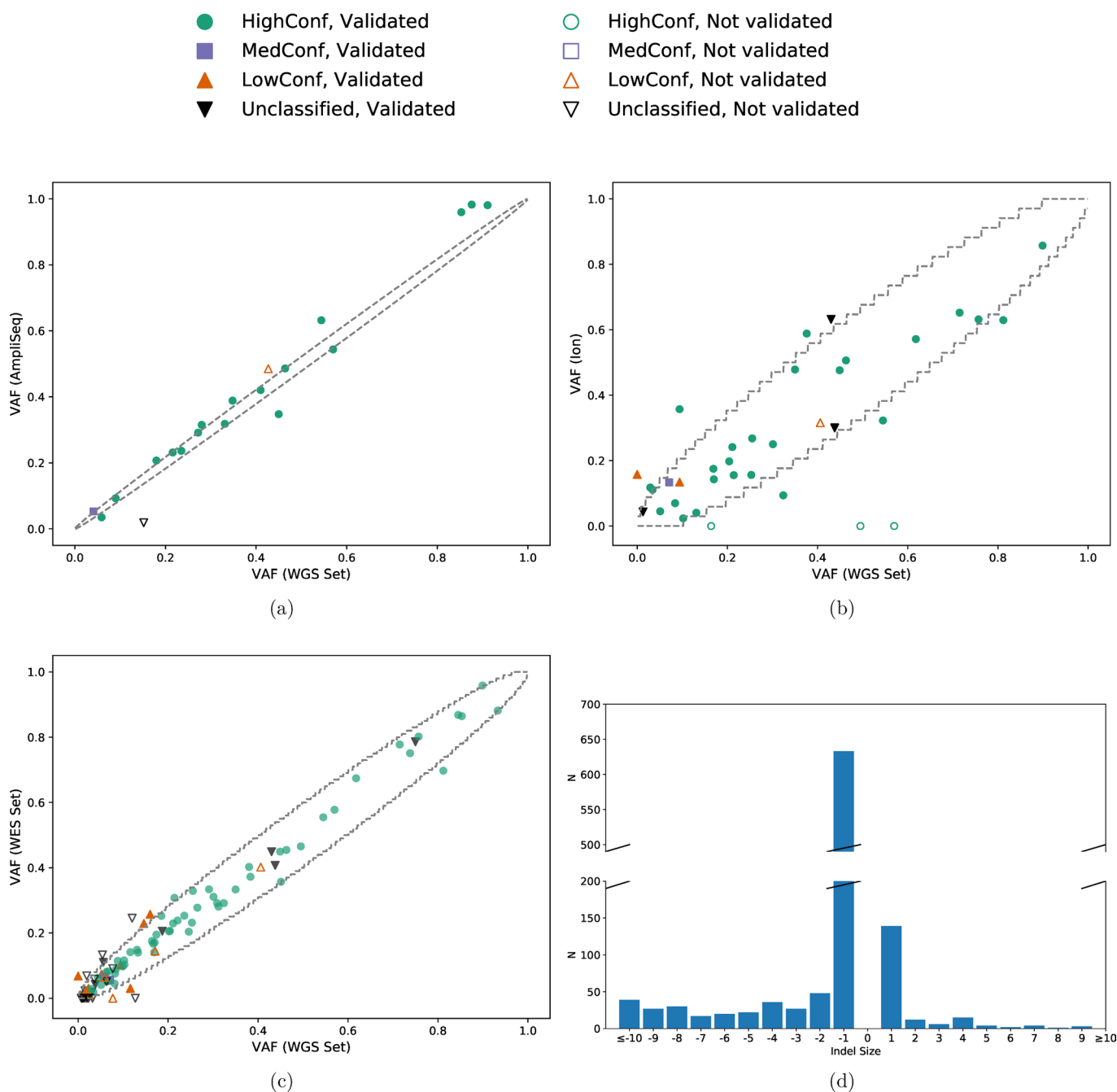
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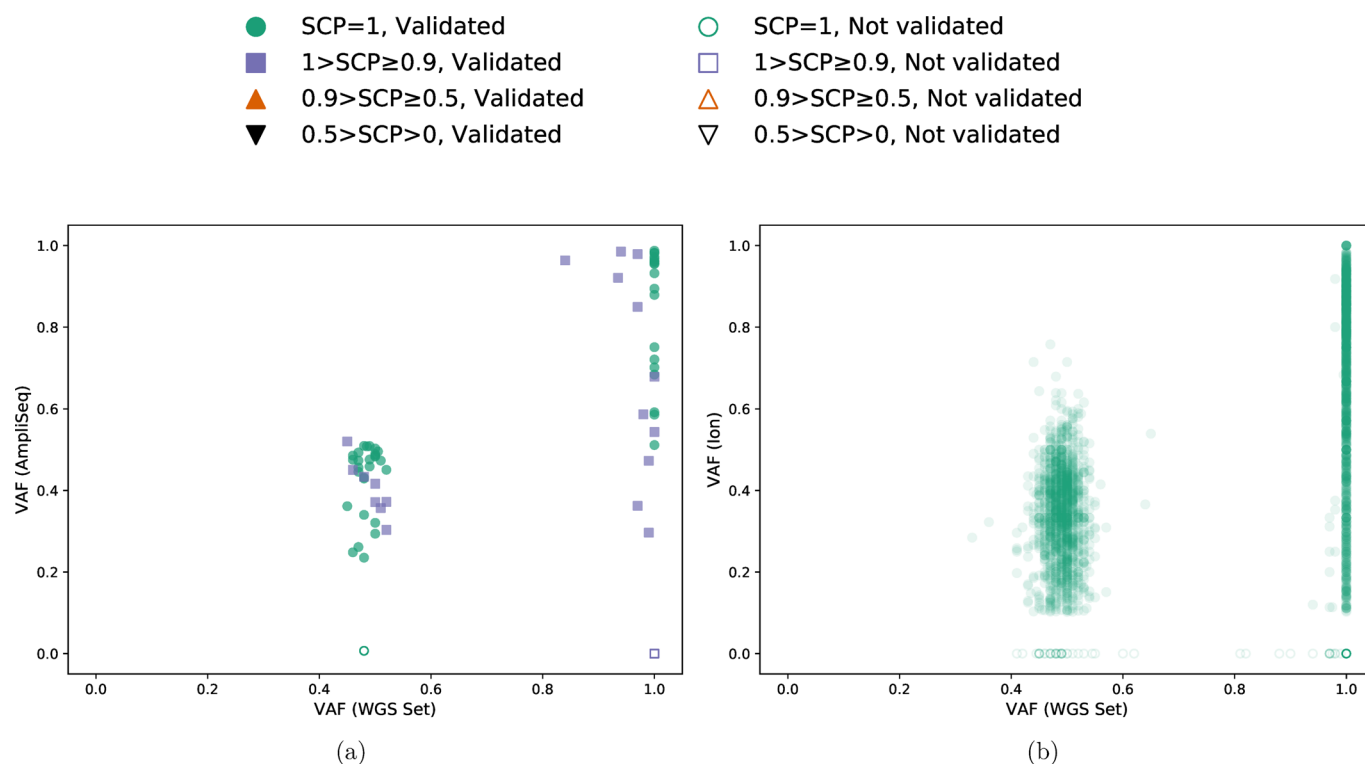
Extended Data Fig. 1 | 3D scatter plot shows the consistency of SomaticSeq and NeuSomatic classification of somatic variant calls. 3D scatter plot for number of PASS classifications by SomaticSeq, NeuSomatic-E, and VAF for **(a)** SNV ($R=0.997$) and **(b)** indel calls ($R=0.925$). **(c)** The subset of SNV calls that were re-sequenced by AmpliSeq. Solid markers are deemed 'validated.' Open markers are deemed 'not validated.' Stars/crosses are deemed uninterpretable. HighConf calls generally have many PASS calls and a full range of VAF. MedConf have fewer PASS calls and tend to have lower VAF. Unclassified calls have a full range of VAF, which means their somatic signals were poor-quality.



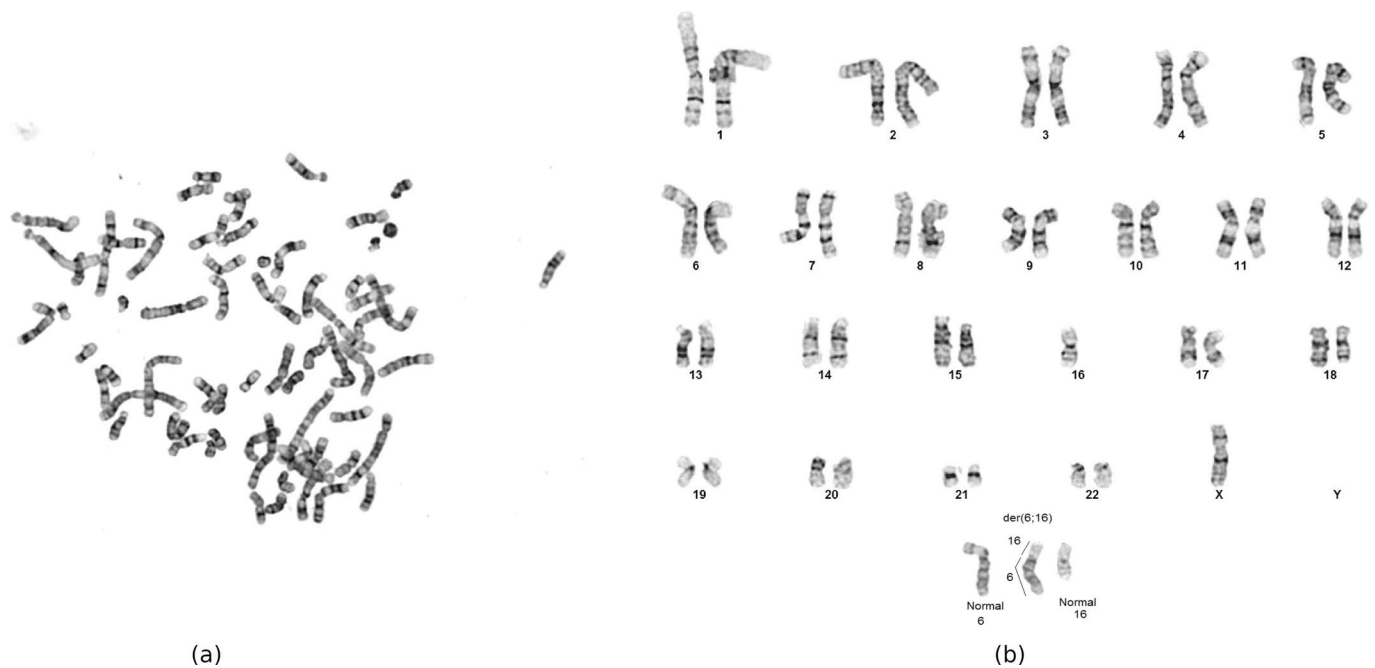
Extended Data Fig. 2 | Genome coverage and high-confidence regions. (a) Genome coverage by reads from three technologies. Inner track: PacBio. Middle track: 10X Genomics. Outer track: Illumina HiSeq. **(b)** Genome regions coverage by short reads in comparison to NA12878. Outer black Track: Gene density plot. Middle orange track: NA12878. Inner blue track: the callable regions in HCC1395.



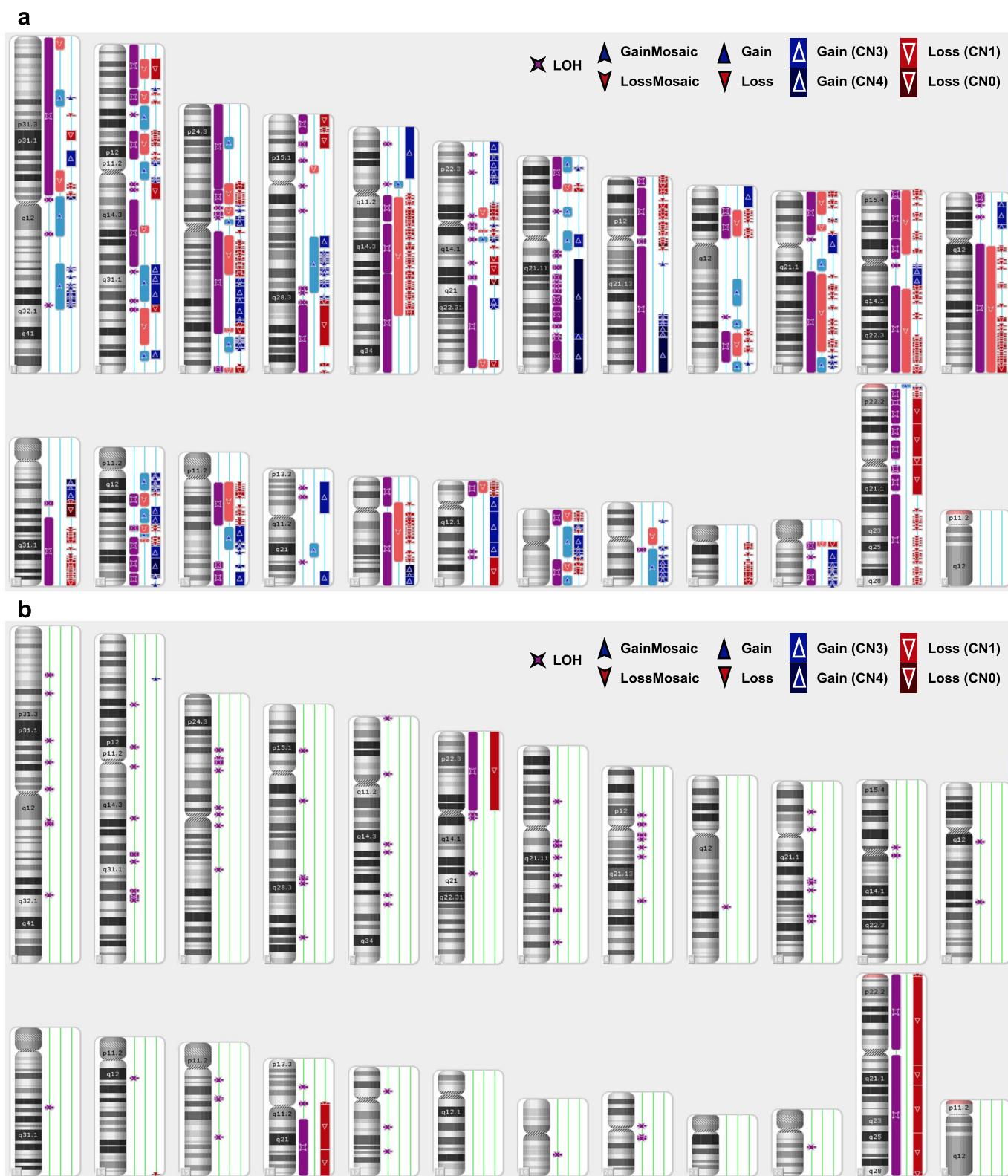
Extended Data Fig. 3 | Validation of somatic indels. (a) Validation of indels by AmpliSeq. R=0.989 for HighConf calls. (b) Validation of indels by WES with Ion Torrent. R=0.767 for HighConf calls. (c) Validation of indels by WES with HiSeq. R=0.990 for HighConf calls. (d) Histogram of indel sizes. The dashed lines on the diagonal for (a), (b), and (c) are the 95% binomial confidence-interval of observed VAF given the actual VAF, calculated based on depths of 2000X for AmpliSeq, 34X for Ion Torrent, and 100X for WES, respectively. (d) shows the indel lengths of the somatic indels in the reference call set.



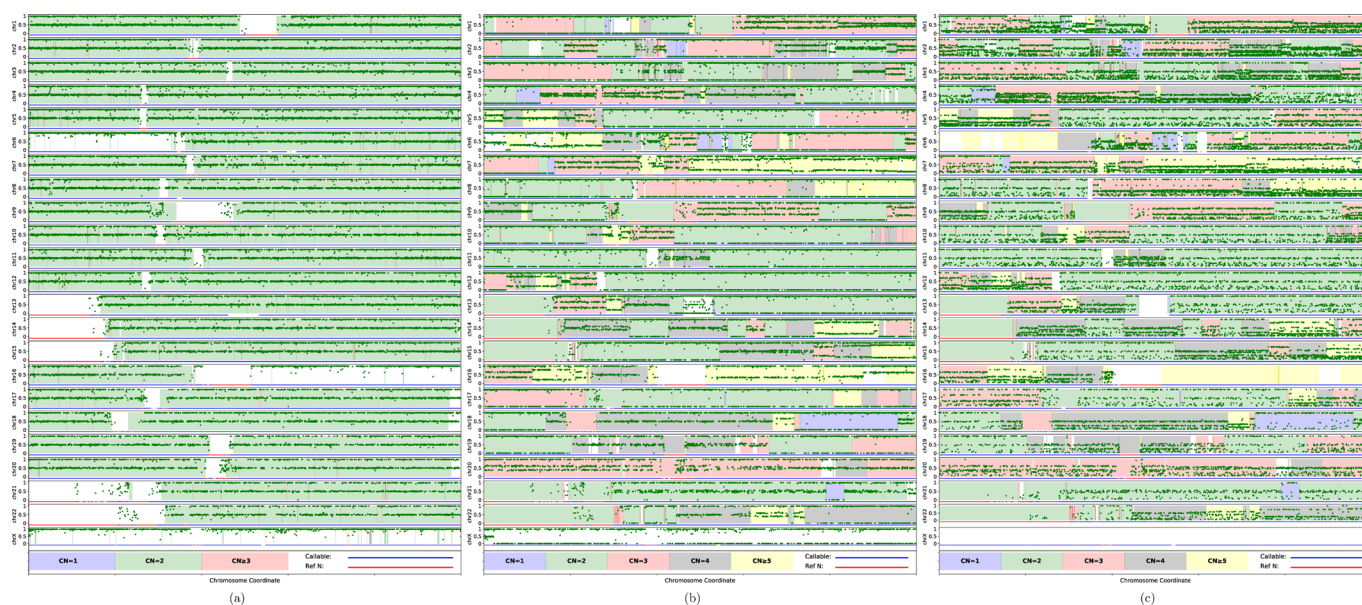
Extended Data Fig. 4 | Validation of germline indels. Germline indel scatter plots comparing VAF super set to confirmed VAF. **(a)** VAF scatter plot of germline indels by WGS super set and AmpliSeq. **(b)** VAF scatter plot of germline indels by truth set and Ion Torrent WES.



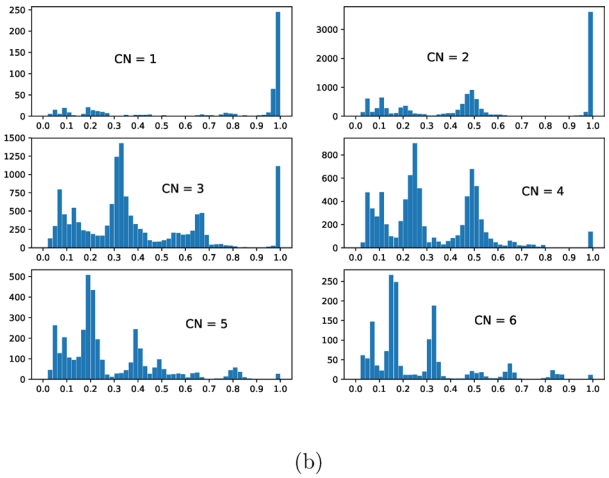
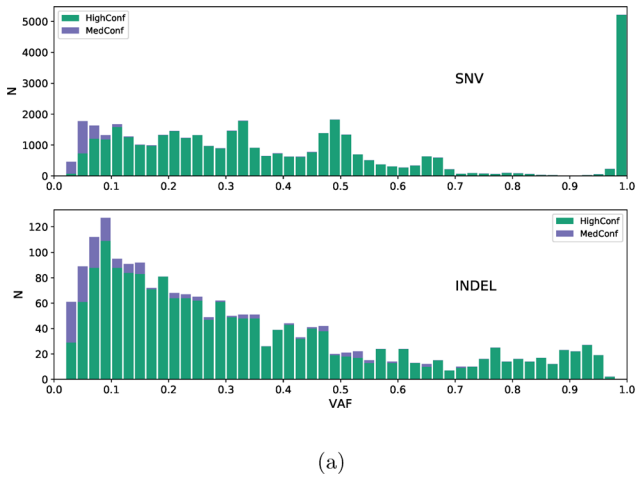
Extended Data Fig. 5 | Karyotyping of HCC1395 and HCC1395BL. (a) Karyotype of HCC1395. Cytogenetic analysis was performed on ten G-Banded metaphase cells from HCC1395. Analysis pointed to a hypertetraploid line with chromosome counts ranging from 64-79 and gain of 38-63 unidentifiable marker chromosomes. **(b)** Karyotype of HCC1395BL. Cytogenetic analysis was performed on ten G-banded metaphase cells from HCC1395BL. All ten cells showed loss of a chrX and an unbalanced whole arm translocation between the long-arm of chr6 at band q10 and the short-arm of chr16 at band p10. This resulted in a net loss of one copy of the short-arm of chr6 and loss of one copy of the long-arm of chr16. The abnormal chromosome could be placed in either a chr6 or chr16 locus as we were unable to determine if the centromere belongs to chr6 or chr16 (inset figure).



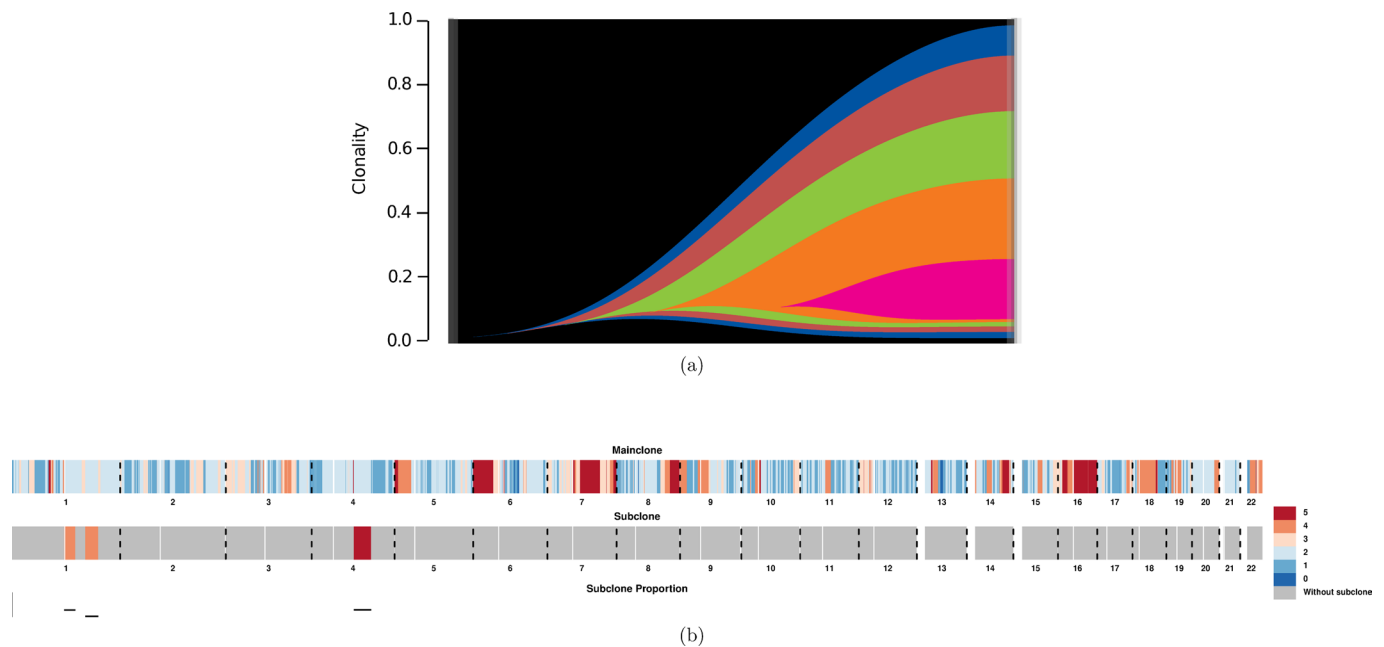
Extended Data Fig. 6 | Cytogenetic analysis with Affymetrix Cytoscan HD microarray. Cytogenetic analysis with Affymetrix Cytoscan HD microarray. **(a)** Cytogenetic view of HCC1395. **(b)** Cytogenetic view of HCC1395BL. The losses of chr6p, chr16q, and chrX were confirmed.



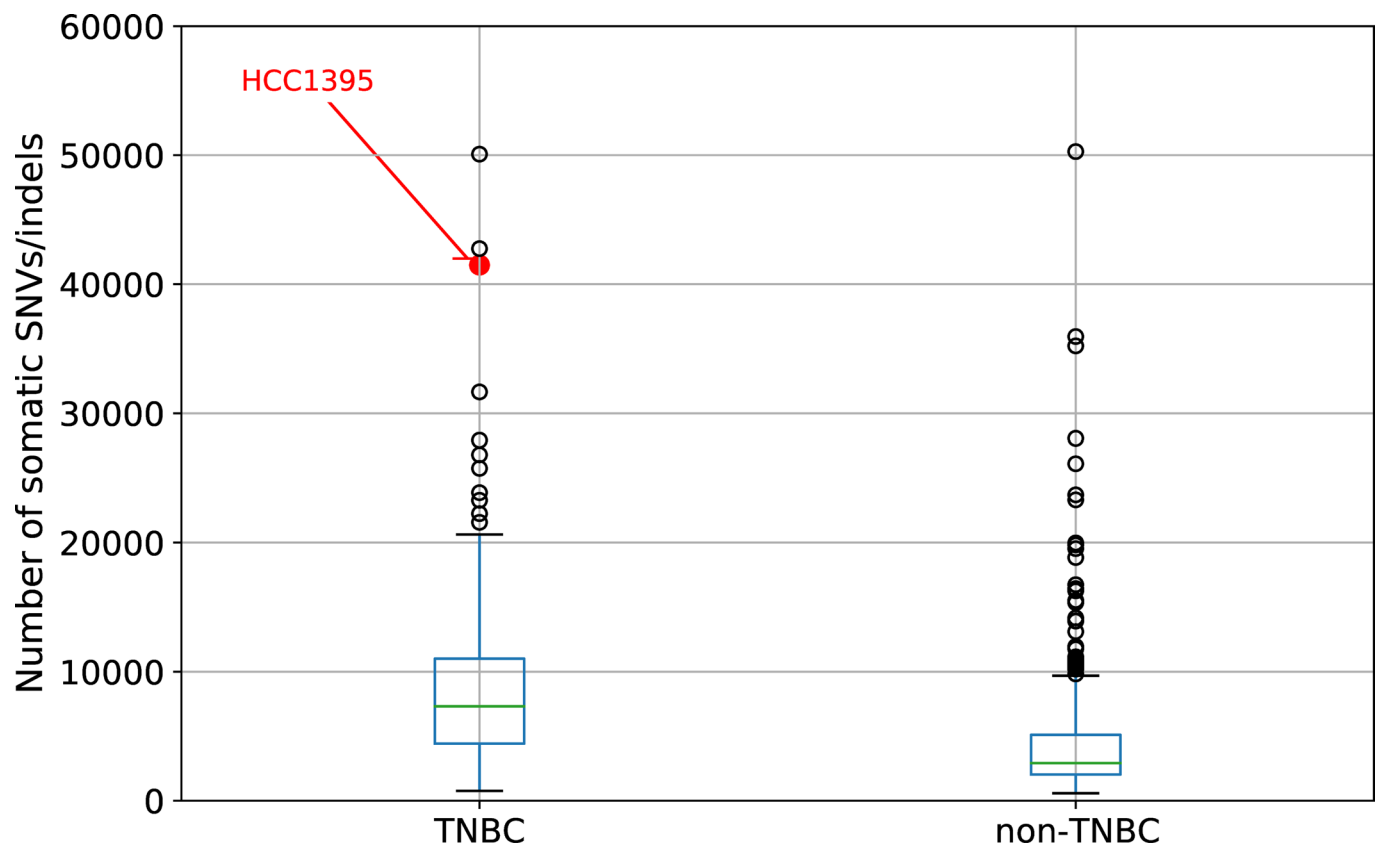
Extended Data Fig. 7 | Variant allele frequencies across the genome. (a) VAF of truth set germline SNVs in HCC1395BL. The copy numbers of HCC1395BL were predicted by Affymetrix Cytoscan HD microarray. **(b)** VAF of the truth set germline SNV positions (discovered in HCC1395BL) in HCC1395. **(c)** VAF of the truth set somatic SNVs in HCC1395. The copy numbers of HCC1395 were predicted by ascatNgs.



Extended Data Fig. 8 | Variant allele frequencies of somatic mutations. (a) VAFs of somatic SNVs and indels in the reference call sets. **(b)** VAFs of reference SNVs in different copy number states as predicted by asc4tNgs.



Extended Data Fig. 9 | Tumor sample HCC1395 CNV and Clonality Analysis. Tumor sample HCC1395 CNV and Clonality Analysis. **(a)** Clonality analysis from WES data using SuperFreq for tumor cell line HCC1395. The clonality of each somatic SNV was calculated based on the VAF, accounting for local copy number. The SNVs and CNAs were evaluated with hierarchical clustering based on the clonality and uncertainty across replicates for HCC1395. The river plot shows the relative distribution of multiple subclones in HCC1395. The main cancer clone (blue) and the two subclones (red and green) appeared in early time of clonal evolution, while subclone (orange) and its descendant (pink) appeared in the late event of the clonal evolution. **(b)** The main- and sub-clonal somatic copy number profiles using subHMM38 from the Illumina WGS data set. Main-clonal genotype: upper panel; sub-clonal genotype: middle panel; sub-clonal proportion: bottom bar plot. Each colored block represents the genotype of somatic copy number alterations (SCNAs) in the corresponding position of the chromosome. The chromosomes are separated by vertical dash lines. Genotype of SCNAs: deletion (DEL), homozygous deletion (HOMD), hemizygous deletion loss of heterozygosity (DLOH), copy neutral loss of heterozygosity (NLOH), diploid heterozygous (HET), gain of one allele (GAIN), amplified loss of heterozygosity (ALOH), allele-specific copy number amplification (ASCNA), balanced copy number amplification (BCNA), and unbalanced copy number amplification (UBCNA).



Extended Data Fig. 10 | Number of somatic mutations detected in HCC1395 and 560 triple negative and non-triple negative breast cancers from previous literature. Number of somatic mutations detected in HCC1395 and 560 triple negative and non-triple negative breast cancers from previous literature⁵⁹.

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n/a Confirmed

- ☐ ☒ The exact sample size (n) for each experimental group/condition, given as a discrete number and unit of measurement
- ☐ ☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
- ☐ ☒ The statistical test(s) used AND whether they are one- or two-sided
Only common tests should be described solely by name; describe more complex techniques in the Methods section.
- ☒ ☐ A description of all covariates tested
- ☒ ☐ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
- ☐ ☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
- ☐ ☒ For null hypothesis testing, the test statistic (e.g. F , t , r) with confidence intervals, effect sizes, degrees of freedom and P value noted
Give P values as exact values whenever suitable.
- ☒ ☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
- ☒ ☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
- ☐ ☒ Estimates of effect sizes (e.g. Cohen's d , Pearson's r), indicating how they were calculated

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection

FASTQ sequence files for WGS and WES were generated from the Illumina sequencer images using the Illumina RTA 1.18.66.3 (HiSeq 2500) or 2.7.7 (HiSeq 4000) and bcl2fastq 2.17.1.14 software; The Illumina sequencer images for the Chromium library sequencing were processed with the Illumina bcl2fastq 2.17.2.1 software and the 10x Genomics Long Ranger 2.1.5 pipelines. Linked-Read data quality was assessed using the 10x Genome browser Loupe; Ion Torrent WES sequencing used the BaseCaller module of the Torrent SuitTM software package v5.8.0 (Thermo Fisher Scientific Inc). Pacbio Sequel sequencing used the Pacbio SMARTlink software v5.0.0. CytoScan array (Affymetrix) was processed with ChAS software (version 3.3)

Data analysis

BAMSurgeon v1.1
Bowtie2 v2.2.9
BWA MEM v0.7.15
DeepVariants v0.5.1
Freebayes v1.2.0
GATK3 v3.8
HaplotypeCaller v3.8.0
MuSE v1.0rc_c
MuTect2 4.beta.6
NovoAlign v3.07.01
RTG v3.8.4
SomaticSeq v2.8.1 / seqc2_v1.2
SomaticSniper v1.0.5.0
Strelka2 v2.8.4
TMAP v3.0.1
TNScope v201711.02
Trimmomatic v0.30
VarDict v1.5.1

NeuSomatic v0.1.3
 AscatNgs v4.2.1
 Cell Ranger DNA 1.1

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- Accession codes, unique identifiers, or web links for publicly available datasets
- A list of figures that have associated raw data
- A description of any restrictions on data availability

All raw data (FASTQ files) are available on NCBI's SRA database (SRP162370). The truth set for somatic mutations in HCC1395, VCF files derived from individual WES and WGS runs, bam files of BWA-MEM alignment and source codes are available on NCBI's ftp site (ftp://ftp-trace.ncbi.nlm.nih.gov/ReferenceSamples/seqc/Somatic_Mutation_WG). Some alignment files (BAM) are also available on Seven Bridges' Cancer Genomics Cloud (CGC) platform under SEQC-II project. Public databases for variant annotation: COSMIC (<https://cancer.sanger.ac.uk/cosmic>), ClinVar (<https://www.ncbi.nlm.nih.gov/clinvar/>)

Field-specific reporting

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Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	DNA extracted from two cell lines (HCC1395 and HCC1395BL) were applied to multiple NGS technologies (HiSeq-WES, HiSeq-WGS, NovaSeq-WGS, PacBio-WGS, 10X Genomics-WGS, Ion Torrent-WES, and NextSeq-Ampliseq). Total of 21 WGS on HiSeq (3x3+3) and NovaSeq (9) with coverage ranging 50X to 100X were used to define initial "truth set" of somatic mutations and germline variants. 12 repeats of HiSeq-WES (3x3+3), a single run of Ion Torrent-WES, and three replicates of NextSeq-Ampliseq were used to confirm three subsets of variants in the "truth set". Results from WGS of PacBio were used to resolve contradicting calls of a small number of variants. Single cell sequencing with genomic DNA (gDNA) was performed on HCC1395 (1465 cells) and HCC1395BL (983 cells).
Data exclusions	All data from planned experiments have been included.
Replication	Six sequencing centers performed 12 repeats (3x3+3) of HiSeq-WGS and HiSeq-WES, and 11 repeats of WGS on the 10X Genomics. A single sequencing center performed 9 repeats of NovaSeq-WGS and 3 repeats of NextSeq-Ampliseq. All attempts at replications were successful.
Randomization	Aliquots DNA from the same lot were randomly distributed to each sequencing center.
Blinding	Since materials derived from two well-established cell lines originated from a consented patient, no blinding was necessary.

Reporting for specific materials, systems and methods

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Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☐	☒ Eukaryotic cell lines
☒	☐ Palaeontology
☒	☐ Animals and other organisms
☒	☐ Human research participants
☒	☐ Clinical data

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Eukaryotic cell lines

Policy information about [cell lines](#)

Cell line source(s)	American Type Culture Collection (ATCC), HCC1395 (CRL2324), HCC1395 BL (CRL2325)
Authentication	The cell lines have been authenticated by STR profile, Karyotyping, PCR Mycoplasma, and Sterility testing.
Mycoplasma contamination	No mycoplasma contamination was detected
Commonly misidentified lines (See ICLAC register)	No commonly misidentified cell lines were used in this study.

Supplementary information

**Establishing community reference
samples, data and call sets for
benchmarking cancer mutation detection
using whole-genome sequencing**

In the format provided by the
authors and unedited

Establishing reference data and call sets for benchmarking cancer mutation detection using whole-genome sequencing

1 Somatic Mutations

1.1 Accuracy and Sensitivity of the Machine Learning Frameworks

The high classification accuracies of both SomaticSeq¹ and NeuSomatic are demonstrated in **Supplementary Fig. 1**. SomaticSeq is an ensemble caller that relies on the six callers (MuTect2², SomaticSniper³, VarDict⁴, MuSE⁵, Strelka2⁶, and TNscope⁷) to produce a combined call set, from which true positives and false positives are distinguished. NeuSomatic is a deep convolutional neural network (CNN) based classification algorithm that has two different ways of generating mutation candidates: 1) the Ensemble mode that relies on the same six callers as SomaticSeq (i.e. NeuSomatic-E), and 2) the Standalone mode that relies on its own internal alignment scanner (i.e. NeuSomatic-S). The sensitivity of SomaticSeq and NeuSomatic-E are limited by the combined call set of the six callers. **Supplementary Fig. 2** shows the VAF of variants that were simulated, detected, and classified correctly. Most undetected mutations have low-VAF. The maximum base call quality value is a Phred score of 40, which translates to an error rate of 1/10,000. While that error rate is impressively low, it is still two orders of magnitude greater than the typical somatic mutation rate in cancer⁸. Thus, even under the most ideal situation and with the best sequencing data, variant calling requires a minimum of two variant reads. Given the ~50X average sequencing depth in this study, that translates to a detection limit of roughly 5% VAF. Thus, only a fraction of simulated variants around or below 5% were detected in each data set, and the observed sensitivity in each data set increased with VAF as expected. The sensitivity of variants in the $2.0\% < \text{VAF} \leq 4.0\%$ range was merely 8.6%. The sensitivity increased to 40.3% for variants of $4.0\% < \text{VAF} \leq 6.0\%$, and only reached 90.9% for variants of $10.0\% < \text{VAF} \leq 12.0\%$. Thus, at this coverage level, low-VAF variants would only be called occasionally, resulting in *initially* low confidence classifications.

1.2 SomaticSeq and NeuSomatic Classifications

SomaticSeq and NeuSomatic machine-learning classifiers were used to score each variant candidate in each data set (**Supplementary Fig. 3a**). For NeuSomatic, both NeuSomatic-E and NeuSomatic-S classifiers were trained. NeuSomatic-E employs the same ensemble of callers to generate variant candidates like SomaticSeq does, whereas NeuSomatic-S scans through the alignments with an internal algorithm to generate variant candidates. NeuSomatic-E has higher accuracy with greater need for computational resources. **Supplementary Fig. 3b and c** show that the breakdown of SomaticSeq and NeuSomatic scores for each variant call in real data is qualitatively similar to those found during cross-validations of simulated mutation data, i.e., vast majority of calls are clear PASS or REJECT, with only a small fraction of calls with intermediate scores (**Supplementary Fig. 1**).

More than 90% of HighConf calls were classified as PASS by SomaticSeq in at least 54 of the 63 data sets (Average 60.5 PASSes and 0.15 REJECT. **Supplementary Fig. 3d**). MedConf calls were classified as PASS by SomaticSeq in fewer data sets (average 27.4 PASSes) but also had limited REJECT classifications by SomaticSeq (average 0.8 REJECT). LowConf calls typically were classified as PASS inconsistently across the 63 data sets (average 8.2 PASSes and 1.7 REJECTs). Unclassified calls were not reproducible and likely false positives, with more REJECT

classifications than PASS (average 1.8 PASSes and 3.8 REJECTs). The majority (82%) of the calls in the all-inclusive set were either HighConf or Unclassified. In other words, all-inclusive set calls were either highly reproducible or extremely irreproducible (**Supplementary Fig. 3d**). The challenge was to confirm MedConf calls and rescue LowConf calls to maximize our reference call set while preventing false positives from mixing in.

The 3-D scatter plots for SomaticSeq and NeuSomatic-E classifications are shown in **Extended Data Fig. 1**. These figures demonstrate the consistency between SomaticSeq and NeuSomatic classifications across all data sets, despite the fact that SomaticSeq and NeuSomatic use totally different machine learning algorithms. For indels, upon deeper investigation, many indel calls that had classification conflicts were long deletions (e.g., ≥ 12 -bp deletions). These long deletions were difficult to detect and classify on a per sample basis due to the lack of reads covering the entirety of the deleted bases. In addition, the presence of soft-clipped reads spanning the deleted regions made them seem low quality. However, when multiple reads spanning long deletions were present, they almost always appeared to be true mutations. Thus, we promoted ≥ 12 -bp deletion calls that were initially labelled as MedConf into HighConf. We have manually inspected over 50 such promotions, and the promotions seemed well justified in each and every case (**Supplementary Data File 3**).

1.3 High-confidence regions

The consensus callable regions consist of a total of 2.73 billion base pairs (bps) (**Extended Data Fig. 2b**). In comparison with GIAB NA12878 genome's high-confidence (HC) regions⁹, 88% of our consensus callable regions are in common with GIAB's HC regions. On the other hand, 98% of GIAB's HC regions are a part of our consensus callable regions. GIAB's HC excluded regions with structural variations as well as regions where variant calls were inconsistent with pedigree or regions with unexplained pipeline inconsistencies. Similarly, we further removed three lost chromosome arms (chr6p, chr16q, and chrX) from the consensus callable regions because somatic and germline variants cannot be distinguished, as well as 20-bp regions spanning LowConf and Unclassified somatic mutation calls where we could not confidently call somatic variants. What has remained is now the high-confidence regions which make up $>82\%$ of the entire genome. We included the SEQC-II high confidence regions accompanying our reference call set. To use the reference call sets, we recommend limiting the evaluation of calling accuracy to the SEQC-II high-confidence regions to ensure maximum reliability. An application of the reference call set is shown in Supplementary Material Sec. 1.9.

1.4 Variant allele frequencies

In **Extended Data Fig. 8a**, it is apparent that there are VAF peaks for SNVs (VAF estimates for indels, especially long indels, are often underestimated because reads with indels near the ends are often soft-clipped. They do not have indel aligned in them, and thus not counted). Most of these VAF peaks correspond to different copy number states, but peaks around and below 15% can be explained by tumor heterogeneity. While HCC1395BL is a relatively unremarkable genome other than the loss of three chromosomal arms, HCC1395 is a highly rearranged near-triploid genome with 66 chromosomes. The VAF of germline variants across the genome mostly resemble a typical genome in HCC1395BL, with the exception of the aforementioned three arm losses where only $\text{VAF} \approx 100\%$ are observed (**Extended Data Fig. 7a**). **Extended Data Fig. 7b**, on the other hand, shows the VAF of the same germline positions in the HCC1395 tumor cell

line. The change in VAF between the same genomic coordinates revealed copy number changes between the normal and tumor genomes. **Extended Data Fig. 7c** shows the VAF of somatic SNVs, which again revealed almost discrete VAFs in different genomic regions with different copy number states; this is consistent with the different VAF peaks shown in **Extended Data Fig. 8b**. In addition, the VAF breaks are also apparent in **Extended Data Fig. 7b and c**. Each implies a structural break in the HCC1395 genome.

We examined the distribution of VAFs and chromosome locations of mutations in all four categories: HighConf, MedConf, LowConf, and Unclassified. While the majority of VAFs in HighConf were between 0.05 and 1, VAFs in MedConf lied between 0.02 and 0.1. Most VAFs in LowConf were less than 0.05. Although most of the VAFs in Unclassified were lower than 0.05, VAFs in Unclassified spanned a full spectrum between 0.01 to 1 (**Extended Data Fig. 1a and b**). Unclassified mutations were most likely artifacts called by a single analysis pipeline or in a single data set and thus were excluded from the reference call set. In addition, over 33% of the mutations in the LowConf and Unclassified groups were outside the consensus callable regions (**Supplementary Table 4**). Therefore, mutations would be classified in the LowConf or Unclassified groups due to: not being called in most of 21 libraries, not being called in every aligner (difficult-to-map regions where the three aligners disagreed), and/or pipeline specific artifacts. The probability of a mutation with ~5% VAF being missed by our all-inclusive set is minuscule (estimated as 2.5×10^{-6} , see Supplementary Material Sec. 1.7). Hence, calling specificity could be evaluated in the SEQC-II high-confidence regions. Specifically, a novel call in the high-confidence region with VAF higher than 5% should be regarded as a false positive.

The majority of somatic indels were insertions or deletions of a single base but ranged from 58-bp for deletions to 14-bp for insertions in the truth set (**Extended Data Fig. 3d**).

1.5 Orthogonal Validations

The overall validation rates described in the main text were calculated based on the number of SNV and indel calls that were validated by at least one of the three validation experiments. While AmpliSeq enrichment was based on randomly selected positions in our super set and only had 7 overlapping SNVs with WES, Ion Torrent and HiSeq WES had many positions in common because they were both whole exome hybrid captures. We have investigated in detail discrepancies between our confidence labels in the super set and any *one* of the three validation experiments. The details for each manually inspected call (i.e., variant coordinates, confidence labels, automated validation determination, and our comments after visually inspecting the alignments with IGV, etc.) are documented in **Supplementary Data File 2**.

To summarize, we have concluded that all the HighConf SNVs “not validated” are in fact real somatic SNVs that were *false negatives in the validation experiments*. Two HighConf indels “not validated” had VAF<0.5% germline signal in 21 WGS normal replicates vs. ~40% somatic signal in the tumor, but we manually relabeled them LowConf to err on the side of precision. All MedConf SNVs and indels were “validated.”

On the other hand, the Unclassified SNVs and indels that were “validated” were mostly due to the detection of variant-supporting reads in the middle of large deletions or SVs (likely false positives), having very messy alignments and many mismatches (likely false positives), having

germline signal in the WGS data that was not captured in low-depth Ion data (likely false positive), being a part of very complex somatic change (possibly true variants we could not confidently classify), or very low VAF (<1.5%) variants (well below our stated LOD of 5%). All of those issues exist for LowConf SNVs and indels that were “validated.”

1.5.1 Orthogonal Confirmation of Call Set with AmpliSeq

We randomly selected 355 SNV and 21 indel calls of different confidence levels from the super set in the consensus callable regions and performed PCR-based AmpliSeq with approximately 2000X depth for tumor and B lymphocyte cells on an Illumina MiSeq sequencer. Our pre-determined rules dictated that positively confirmed calls must have at least 100 variant-supporting reads in the tumor, and no variant read in the normal with sequencing depths of $\geq 600X$ in the normal. Not confirmed calls had fewer variant-supporting reads than the expected base call errors, and/or had $VAF \geq 10\%$ in the matched normal (germline variant false positives). Uninterpretable calls did not satisfy the criteria for either positive or no validation because they did not have enough read depth (<50) in the tumor and/or normal (See Methods for details).

Both HighConf and MedConf SNV calls had very high average coverage in validation and thus had impressive validation rates (99.6% and 100%, respectively) (**Table 2**). There was one single HighConf SNV (chr3: 71328321 C>T) that was not confirmed by AmpliSeq. In AmpliSeq data, there were zero variant-supporting reads and zero forwardly-oriented reads, indicating issues with PCR enrichment, sequencing, or mapping. However, this variant was called PASS by SomaticSeq, NeuSomatic-E, and NeuSomatic-S in 62 out of 63 WGS data sets. In addition, confirming reads were also found in PacBio data as well. Therefore, based on overwhelming support in the WGS data sets in addition to PacBio data sets, we concluded that it was “not validated” due to *false negative of AmpliSeq*, and *not* due to error of our HighConf label. The six uninterpretable HighConf calls are also documented in **Supplementary Data File 2**.

There were also four Unclassified calls that were “validated,” as shown in **Supplementary Table 6**. The calls at chr3:117937789 (**Supplementary Fig. 4a**) and chr11:24722179 appeared 1-bp away to 13-bp deletions. Adjacent indel is a known false positive marker for SNV calls because it may be a result of misalignment. In addition, indels, as large as 13 bp, create a large edit distance in the alignment, which is one of the top false positive markers in SomaticSeq. The vast majority of calls with large edit distances are false positives, hence SNVs next to large indels present a particular challenge as the SNV is masked among false positive signals. We are not particularly certain if the two cases presented here are actual mutations or false positives, though we’ve kept their “confirmed” status in the validation rate calculation. The other two “confirmed” Unclassified calls are in the middle of a 38-bp deletion (**Supplementary Fig. 4b**), and very much look like false positives of AmpliSeq. Thus, while AmpliSeq has “confirmed” four out of 22 Unclassified calls, two of them are possibly false positives and the other two are almost certainly false positives. Therefore, the “validation rate” of 18.2% may be an overestimate of false negative rate in Unclassified calls.

The VAF calculated from the WGS data sets correlated highly with the VAF calculated from the AmpliSeq data set for HighConf calls ($R=0.982$, **Fig. 2d**). On the other hand, almost all the data points at the bottom of the graph (i.e., $VAF = 0$ from AmpliSeq data) are Unclassified calls (black). This means some Unclassified calls have high VAF calculated from 21 WGS data sets, but they

were categorized correctly as Unclassified (meaning likely false positives). In addition, a large number of uninterpretable Unclassified calls (black 3-pointed stars) lying at the bottom suggest those were correctly labeled as Unclassified in addition to the not confirmed ones (open black triangles). Moreover, the four “validated” Unclassified calls had dubious supporting evidence as well. Taken together, these results suggest that the actual true positive rate for the Unclassified calls may be even lower than the validation rate (18.2%) we reported here.

The indel equivalent is portrayed in (**Extended Data Fig. 3a**). There were 18 HighConf, 1 MedConf, and 2 Unclassified indel calls re-sequenced by AmpliSeq. The only HighConf indel call (chr1:111980493 GA>G) not confirmed was caused by discovery of germline signal in the matched normal in AmpliSeq data, whereas there was only one single variant-supporting reads in all 21 normal WGS data combined (1 out of 788 total reads). We could not make a definitive conclusion based on PacBio data, either. Thus, we manually re-labeled this indel call from HighConf to LowConf in this release. The lone Unclassified indel call was not confirmed (we expect Unclassified calls not to be confirmed).

1.5.2 Orthogonal Confirmation of Call Set with WES on Ion Torrent

We have also sequenced the tumor-normal pair with whole exome sequencing (WES) on the Ion Torrent S5 XL sequencer with the Agilent SureSelect All Exon + UTR v6 hybrid capture. The sequencing depths for the HCC1395 and HCC1395BL were 34× and 47×, respectively. Results from this Ion Torrent sequencing were leveraged to evaluate higher-VAF SNV calls due to its relatively low coverage (**Table 1 and 2**). HighConf and MedConf SNV calls had high positive validation rates (99.1% and 100%). However, because Ion Torrent sequencing was performed at much lower depth, only 54% of the calls were deemed interpretable (compared with 91.5% for AmpliSeq). The trend of higher uninterpretable fraction with lower confidence level calls was also more pronounced in this data set because the coverage was too low to interpret many low-VAF calls.

Upon further investigation of the six HighConf SNVs and four HighConf indels “not validated” by Ion Torrent did indeed have variant-supporting reads in tumor and reference-reads only in normal, thus consistent with somatic mutation status. However, reads in those positions did not meet our pre-determined minimum mapping quality rules, so the variant-supporting reads did not count, resulting in their “not validated” determination. Based on this evidence, we had little reason to “overturn” the HighConf labels for these variant calls.

On the other hand, many of the “validated” Unclassified calls are in low-complexity regions and often have unexplained differences between tumor and normal in the vicinity of the variant calls. While some of them can be real mutations, we cannot currently resolve their status. For the three Unclassified “indels” that were validated, 1) chr10:82232948 TG>T is a few bases away from a 10-bp deletion, so if true it is a complex variant; 2) chr15:30951922 T>TG occurs right next to a SNV, so if true it is also a complex variant; and 3) chr20:63693059 has a 22-bp deletion that has a VAF of 1.4% (called by SomaticSeq 5 times, NueSomatic-E once, and NeuSomatic-S 0 time.), but got captured in the Ion Torrent data. This is probably a true mutation but its VAF is below our LOD. Like all LowConf and Unclassified calls, those positions were removed from our high-confidence regions.

The VAF correlation for HighConf calls between WGS data sets and Ion Torrent WES ($R=0.930$) was lower than that between WGS and AmpliSeq, likely due to low coverage ($34\times$), although the vast majority of the HighConf SNV calls in Ion Torrent data still stayed well within the 95% confidence interval (**Fig. 2e**). There were more uninterpretable Unclassified calls due to possible platform-specific bias (black 3-point stars) at the bottom as well, which is suggestive that the true positive rate for Unclassified calls may be lower than the reported validation rate (15.5%) for Ion Torrent data as well. The indel equivalent is included in **Extended Data Fig. 3b**.

1.5.3 Independent Confirmation of Call Set with WES on HiSeq

We used 16 HiSeq WES replicates from six sequencing centers to evaluate the concordance between these data sets and the WGS data sets employed to construct the super set. While the WES data sets were not sequenced from orthogonal platforms, they provide insights in terms of the reproducibility of our call sets in different library preparations and sequencing strategies. The scatter plots between the super set derived VAF and medium HiSeq WES-derived VAF is presented in **Fig. 2f**. All variants in the reference call set (HighConf and MedConf calls) variants had consistent VAF calculated from both sources.

Again, simple rules were implemented for validation with the WES data as well (**Table 2**). The validation rate for HighConf, MedConf, LowConf, and Unclassified SNV calls by WES were 100%, 100%, 83.5%, and 13.4%, consistent with the previous two methods. The validation rate for HighConf, MedConf, LowConf, and Unclassified indel calls by WES were 98.2%, 100%, 76.9%, and 38.5%. The high-confidence variants were completely reproducible in WES. The one HighConf indel that was “not validated” by WES had a little bit of germline signal: a total of 5-variant supporting reads, 833 in total from 21 WGS replicates, but higher in some WES sets so it was “not validated.” It occurred in a homopolymer region (11A’s), so that it is possible to have indel sequencing error in the normal at 0.6%, against VAF in the tumor of 41% (251/614). This is probably a real indel. Nevertheless, we also removed it from the reference call set as a LowConf because we err on the side minimizing false positives.

The large number of uninterpretable Unclassified calls at the bottom (black 3-pointed stars) again suggests that the actual true mutation rate among Unclassified calls may indeed be lower than the reported validation rates. The indel equivalent is the subject of **Extended Data Fig. 3c**.

1.6 Overlapping positions between AmpliSeq Target and Ion Torrent WES

There were only seven variant calls that were re-sequenced by both AmpliSeq and Ion Torrent (**Supplementary Table 7**). We highlighted the one call where the validation interpretations were different. The HighConf call at chr15:28815512 was deemed “not confirmed” by Ion Torrent because all the variant-supporting reads had low mapping quality scores only in Ion Torrent data and thus were not counted. It was one of the six “not confirmed” HighConf SNV calls by Ion Torrent that indeed seemed like true positives, and it was “confirmed” by AmpliSeq.

1.7 Rescue low-VAF variant calls

The 21 WGS tumor-normal data sets we relied on to discover somatic mutations had average sequencing depths around 50X. Thus, $VAF < 5\%$ variants were not expected to be highly reproducible and would be “lucky” to make it as a HighConf call in our *initial* definition. We estimated based on binomial distribution the probability that a $VAF=5\%$ variant would be

captured as a HighConf mutation using only the 21 50X WGS replicates. With the assumption that three variant reads are necessary and sufficient to make a somatic mutation call (in reality two reads are necessary and can be sufficient, but sufficiency also depends on mapping, alignment, and sequencing qualities that are impossible to estimate accurately), the probability that a VAF=5% variant will be captured in the majority of the replicates and the majority of the sequencing centers (hence a HighConf label) is just 37%. However, the probability that it will escape the all-inclusive call set entirely (one out of 21 replicates will make the super set) is close to zero ($\sim 2.5 \times 10^{-6}$). In other words, while a low-VAF variant will unlikely be labeled as a high-confidence somatic mutation initially, it is unlikely to escape the all-inclusive call set entirely.

To mitigate the shortcoming of 50X sequencing data, we used additional 400X and 350X data sets to promote the confidence level classifications of low-VAF variants in our initial call set, and to rescue previously undetected variants. With the same set of assumptions, even at half of the average depths (i.e., 200X and 175X), a VAF=5% will be captured by both 400X and 350X data sets (and therefore promoted into the high-confidence call set) at 99.4% probability. On the other hand, VAF=3% variants are expected to be “rescued” at only 84.6% probability. Therefore, the limit of detection (LOD) of our reference call set is about 5%. We hereby promoted 1922 SNVs and 98 indels into the high-confidence reference call set at this step. We additionally added 554 LowConf SNVs and 79 LowConf indels into the all-inclusive set, as none of them had consistent SomaticSeq classifications across either the two samples or the three aligners.

Supplementary Fig. 2b shows the VAF for the combined NovaSeq simulation, where 5/9 of the replicates were designated as tumor and 4/9 of the replicates as normal, to create an average sequencing depth of 203X for the designated tumor and 160X for the designated normal. The sensitivities for variants with $2.0\% < \text{VAF} \leq 4.0\%$ and $4.0\% < \text{VAF} \leq 6.0\%$ were 78.4% and 90.3% respectively, a vast improvement from the 50X training set where the sensitivities were 8.6% and 40.3%, respectively (Supplementary Section. 1.1). It stands to reason that the sensitivities for low-VAF variants in 400X and 350X sequencing data are higher.

Finally, to further maximize our sensitivity for low-VAF variants, we created a pair of tumor-normal 1500X WGS data sets by combining all of the short-read WGS data sets into one. We used three aligners and NeuSomatic-S to call variants in them. Because we expected that the 400X and 350X data sets would have captured almost all of the $\text{VAF} \geq 5\%$ variants already, we set up a rather conservative criterion here to promote additional calls into the high-confidence reference call set. We only promoted calls into the high-confidence call set if in the data sets where the variant was not called, there were ≤ 3 reads. In other words, if a call was LowConf or Unclassified previously due to mapping or alignment, we would not promote them here. We only promoted calls that were LowConf or Unclassified due to low signal. We additionally promoted 915 SNVs and 34 indels into the high-confidence reference call set at this step. We have also added 9 more LowConf SNVs and 6 more LowConf indels that were previously undetected into the super set.

While our binomial estimates did not prove that we have not missed any low-VAF somatic variants, it has shown that the chances of low-VAF false negative due to low sequencing depths are very low due to our use of 21 independent replicates, and then additionally 350X, 400X, and 1500X data sets. The binomial estimates can be found in our code repository.

1.8 Undetected *in silico* variants

We investigated the small number of high-VAF ($\geq 40\%$) *in silico* variants that were not detected by any caller, and thus undetectable by SomaticSeq and NeuSomatic-E as well. We found the reasons preventing detection to be a combination of low coverage and poor mapping (**Supplementary Fig. 5**). The majority of the undetected variants had coverage of fewer than 10 reads (**Supplementary Fig. 5a**), and the vast majority of them had ≤ 1 variant-supporting read (**Supplementary Fig. 5b**). Moreover, these regions tended to have very low mapping quality scores (**Supplementary Fig. 5c**) and with many MQ=0 reads, **Supplementary Fig. 5d**. The mapping qualities were so low for these reads that when BAMSurgeon mutated a base in those reads, they no longer mapped to their original positions. That change in alignment was the reason for zero variant-supporting read in many of these *in silico* mutations. In cases like these, the original mapping was also questionable. Low mappability regions were excluded from our high-confidence regions in this release. These low mappability regions represent limitations of today's short-read sequencing technologies. It also presents opportunities for the development of next-generation technologies, as long-read sequencing technology achieves higher accuracy.

1.9 How to use the truth set to benchmark a call set

Here, we demonstrate an example of how to use our reference call set to evaluate the accuracy of a pipeline. We used the hap.py software recommended by the Global Alliance for Genomics and Health (GA4GH)⁹ to benchmark the Strelka2 call set generated from the tumor BAM file with the highest sequencing depth (87X): NV_T_2 vs. NV_N_2 aligned with NovoAlign.

```
hap.py/bin/som.py -N -o /data/PATH/benchmark_results -r /data/PATH/GRCh38.fa -R /data/PATH/High-Confidence_Regions.bed /data/PATH/sSNV.MSDUKT.superSet.v1.2.vcf.gz /data/PATH/WGS.novo.dedup-NV_T_2_vs_NV_N_2-Strelka.snv.vcf.gz
```

Of the 63 Strelka2 call sets, this call set had the highest sensitivity at nearly 97.2% (1,083 false negatives), which is not surprising since it was generated from the data set with the highest sequencing depth. It had a precision of 86.8% (5,794 false positives).

The false negatives are heavily biased toward MedConf calls. While MedConf calls only consist of 5.5% of the total SNVs in the reference call sets, nearly 60% of the false negatives here are MedConf variants. This is not surprising, since MedConf calls are almost exclusively low-VAF variants that are not reproducible in these 50X WGS data sets.

To show that our reference call set correctly included these MedConf calls into the reference call set, we visually inspected 10 of them on IGV using the 350X tumor-normal data to make sure that they were real mutations that were missed in this Strelka2 call set on NV_T_2 vs. NV_N_2. Indeed, all of them look like real mutations, with VAF ranging between 3% and 7%. In the NV_T_2 replicate on which Strelka2 ran, however, the total number of variant-supporting reads for these calls range from 0 to 3. Thus, it is not surprising that those variants were not detected in NV_T_2. An example of such a false negative is shown in **Supplementary Fig. 6a**.

Conversely, we also randomly inspected 10 false positives to make sure that our reference call set did not miss them, and all of those 10 calls were clearly not real mutations. They either had clear germline signals in the 350X SPP data, or very messy alignments that clearly indicated that they were false positives. Two representative examples are shown in **Supplementary Fig. 6b** for

messy alignments and **Supplementary Fig. 6c** for germline false positive. The 10 false negatives and 10 false positives are documented in the **Supplementary Data File 1**.

1.10 Additional functional features and tumor mutation burdens

Tumor mutational burden (TMB) is defined as the number of non-synonymous somatic mutations per Mbps⁴⁴. Recent literature has increasingly reported the correlation between TMB and response to anti-PD(L)-1 immunotherapy⁴⁵. The “gold standard” of measuring TMB is to perform tumor-normal WES and find the total number of non-synonymous mutations (in the coding regions). Due to the higher cost and more time required for WES, researchers are trying to infer TMB using less expensive targeted oncology panels with a small number of driver genes. One way to increase the statistical efficiency of estimating TMB for a targeted panel is to use all somatic mutations, including synonymous mutations which are expected to correlate highly with frequency of non-synonymous mutations if assuming most non-synonymous mutations, especially in high-TMB patients, are not enriched in driver genes compared with synonymous mutations. We inferred TMB with various commercially available target panels. The uncertainties of mutation rate (calculated as the 95% binomial confidence interval) inferred by smaller oncology panels are quite large, so we advise caution when attempting to infer TMB from targeted oncology panels (**Supplementary Table 5**).

2 Germline variants

2.1 Indel validations

The indel equivalent for germline SNV validation with AmpliSeq and Ion WES (shown in the main text **Fig. 4**) is shown in **Extended Data Fig. 4**.

3 Step-by-step procedures to determine the confidence-level of somatic mutations

Here, we document the procedures and commands that defined the four evidence levels (i.e., HighConf, MedConf, LowConf, and Unclassified) in the somatic mutation all-inclusive call set. Instructions to reproduce our results and the location of the files are also documented and will be kept up-to-date at the Somatic Mutation Working Group’s website: <https://sites.google.com/view/seqc2>

3.1 Creating semi-synthetic BAM files as training data

We have multiple intra-center sequencing replicates in 12 Data Groups: HiSeq @IL, HiSeq @FD, HiSeq @NV, and NovaSeq @IL, each aligned with three aligners (BWA MEM¹⁰, Bowtie2¹¹, and NovoAlign¹²). We employed BAMSurgeon to spike *in silico* SNVs and indels into one of the normal replicates as the *in silico* tumor. These *in silico* tumor-normal pairs served as training data to create SomaticSeq and NeuSomatic classifiers.

An example command to create an *in silico* tumor-normal pair is as follows:

```
somaticseq-2.8.1/utilities/dockered_pipelines/bamSimulator/BamSimulator_multiThreads.sh --output-dir /ABSOLUTE/PATH/bwa.NV1N_vs_NV2N --normal-bam-in /ABSOLUTE/PATH/WGS_NV_N_1.bwa.dedup.bam --tumor-bam-in /ABSOLUTE/PATH/WGS_NV_N_2.bwa.dedup.bam --normal-bam-out normalDesignate1N.bam --tumor-bam-out tumorDesignate2N.bam --genome-reference /ABSOLUTE/PATH/hg38.fa --num-snvs 100000 --num-indels 20000 --num-svs 1000 --min-vaf 0 --max-vaf 1 --left-beta 2 --right-beta 5 --max-depth 1000 --seed 1981 --threads 36 --merge-output-bams
```

We also had two additional deep sequencing WGS data sets: 1) 400X NovaSeq data set created by combining the nine sequencing replicates from NovaSeq @IL, and 2) 350X tumor-normal WGS data sets from GT as a part of the tumor-normal titration series. For each of these two data sets, the same three aligners were used.

The training data sets for the 400X NovaSeq were created by combining 5/9 normal replicates as the pseudo-tumor, and the 4/9 remaining normal replicates as the normal. *In silico* mutations were then spiked into the designated tumor. Two such training data sets were created. The sequencing depth of the training data were about 220X for the *in silico* tumor and 180X for the designated normal. The classifiers are still accurate for 400X depth data (**Supplementary Table 8**).

Supplementary Table 8 shows that using the classifiers created by single NovaSeq replicates, such that their sequencing depths were about 1/4 the target data (training set described in previous paragraph), the accuracies were equivalent to the cross-validation results for the 12 Data Groups (**Supplementary Fig. 1**), even though the cross validation of the two training sets with equivalent depths displayed higher accuracies. This improvement in accuracy could be attributed almost exclusively to low-VAF variants (**Supplementary Fig. 2**). The training sets created for combined NovaSeq were about 1/2 the target data, so the discrepancy in sequencing depth is less than those shown in **Supplementary Table 8**. The training data for the 350X WGS data were created by randomly splitting the normal BAM files into a designated tumor and normal at a 55:45 ratio.

3.2 Building SomaticSeq and NeuSomatic classifiers

For each of the 12 Data Groups with sequencing replicates, two training data sets were created as described in Sec. 3.1. For each Data Group, the two training data sets were combined to create SomaticSeq classifiers specific to that Data Group. Additionally, all the training data sets were combined to create SomaticSeq (for LL, EA, and NC) and NeuSomatic classifiers (for all WGS data sets).

3.3 Classification of each pair of tumor-normal BAM files

For each of the 63 tumor-normal pairs, we have run SomaticSeq, NeuSomatic-Ensemble (E), and NeuSomatic-Standalone (S) pipelines.

3.4 Scoring each Data Group (for each of the three aligners)

The steps in Sec. 3.4 - 3.9 describe in detail the schematics shown in main text's **Fig. 1**, i.e., the algorithm that determined the four confidence levels. Sec. 3.4 – 3.7 rely on SomaticSeq and made up the *initial* definition of our call set. In Sec. 3.8 – 3.9, NeuSomatic classifications of the 63 WGS data sets plus the 1,500X pooled WGS data sets were taken into account to make up our final definition of this v1.2 release. The commands used can be found in our code repository.

There were five Data Groups for each aligner, resulting in 15 Data Groups as shown in main **Fig. 1**. In each Data Group, a somatic mutation candidate got a +1 score if it was classified PASS in a sample, a -1 if it was classified REJECT in a sample, and 0 otherwise (LowQual or undetected). The sample scores were summed up in each Data Group. Then, Data Group scores of +3, +1, 0, or -3 were assigned based on the two tables on the left of **Supplementary Fig. 7**. The upper left

table shows all the possible combinations of sample scores for Data Groups with three sequencing replicates, i.e., to get a Data Group score of +3, at minimum two out of three samples had to be PASS, without a single REJECT. While a Data Group score of +1 was not as confident as +3, it still showed positive supporting evidence. A Data Group score of -3 is a “mirror image” of +3, having considerably more contradicting evidence than supporting evidence. The lower left table shows all the possible combinations for the 9-replicate Data Groups, i.e., NovaSeq @IL.

3.5 Aligner-Centric Classifications

For each aligner, the Data Group scores for the five Data Groups were summed together to make an aligner-centric classification. The right table of **Supplementary Fig. 7** shows every possible combination of the five Data Group scores, and the aligner-centric classifications associated with each combination.

3.6 Assigning the initial four Confidence Levels

First, we assigned initial tiers based strictly on the three aligner-centric classifications.

Supplementary Table 9 shows the criteria and the number of total calls that fell into each tier.

The initial tiers provided a starting point to determine how confident we were that a call was a true mutation. In addition, we also took into account certain “red flags” (i.e., high number of MQ0 reads, germline signal, high edit distances, VAF inconsistent with GT’s tumor-normal titration data sets, or a complete lack of a supporting call from one of the three aligners) to create the four confidence levels of HighConf, MedConf, LowConf, and Unclassified.

The algorithm to initially assign such confidence levels is at SomaticSeq’s public seqc2 branch in Github. A brief description is as follows. First, for each aligner, we established a low-end MQ and a high-end edit distance (BAM tag: NM) threshold with the bottom/top 2% of values among calls that were classified as PASS every single time. Then, we used the following rules to assign the four confidence levels:

1. Tier 1 calls defaulted to HighConf, unless the call was not in the consensus callable region, or if there were more MQ0 reads than there were samples, in which case they got demoted to MedConf.
2. Tier 2 calls also defaulted to HighConf, but there were more opportunities to be demoted to lower confidence-levels, such as zero PASS calls from the third aligner, presence of germline signal, location outside the consensus callable region, VAF not moving consistently with the tumor-normal titration, or if the MQ or NM exceeded the established thresholds.
3. Tier 3 calls defaulted to MedConf and could be demoted to lower confidence-levels by the same things that demoted Tier 2 calls.
4. Tier 4 calls defaulted to Unclassified but could be promoted to LowConf if they had more sequencing replicates where the call was classified as PASS vs. REJECT.

3.7 Rescue low-VAF and lower confidence level calls

As described previously, for low-VAF ($\leq 15\%$) calls with LowConf and Unclassified labels, two deep sequencing data sets were used to rescue those calls: 1) nine NovaSeq data sets combined into 400X, and 2) a 350X tumor-normal WGS data set that was a part of a tumor-normal titration

series. If the calls were deemed PASS in those data sets, they would be promoted into the high-confidence call set.

3.8 Modify confidence labels with NeuSomatic classifications

For each variant call in the super set, we have annotated in how many data sets they were classified PASS by both NeuSomatic modes, i.e., Ensemble mode (NeuSomatic-E) and Standalone mode (NeuSomatic-S). We have examined and re-labeled the confidence of calls when the number of data sets classified as PASS by SomaticSeq deviated significantly ($p < 0.05$, two-proportion z-test) from that of NeuSomatic-E/NeuSomatic-S. We also added an additional 9 LowConf SNVs and 6 LowConf indels into the super set at this step. These calls were previously undetected by SomaticSeq.

3.9 Use 1500X call sets for final rescue of low-VAF calls

We combined all short-read WGS data sets into a pair of combined tumor-normal data sets of 1500X in depth, for each of the three aligners. We used these data sets to promote low-VAF low-signal calls into the high-confidence call set (i.e., if the variant call was rejected in 50X data sets due to low signal).

3.10 Use PacBio data to remove inconsistent variant calls

Our PacBio data's sequencing depth is about 30-40X, and also has error rates higher than many variant calls in the reference call set (**Suppl. Table 2**). Thus, PacBio was not used for variant discovery, but we used it to remove variants from low-mappability regions (i.e., PacBio MQ < 50, short-read MQ0 reads > 10, or Phred-scaled linguistic sequence complexity¹³ < 12) from the reference call set if 1) PacBio has no variant-supporting read, and 2) one-sided z-test yields a p-value < 0.05.

This is the final step in our establishment of the v1.2 reference call set. Variant calls that are high confidence as true positives in the High-Confidence Regions are labeled PASS in the FILTER column of the VCF files.

4 Software and Version

Software information is presented **Supplementary Table 10**.

5 Statistical methods

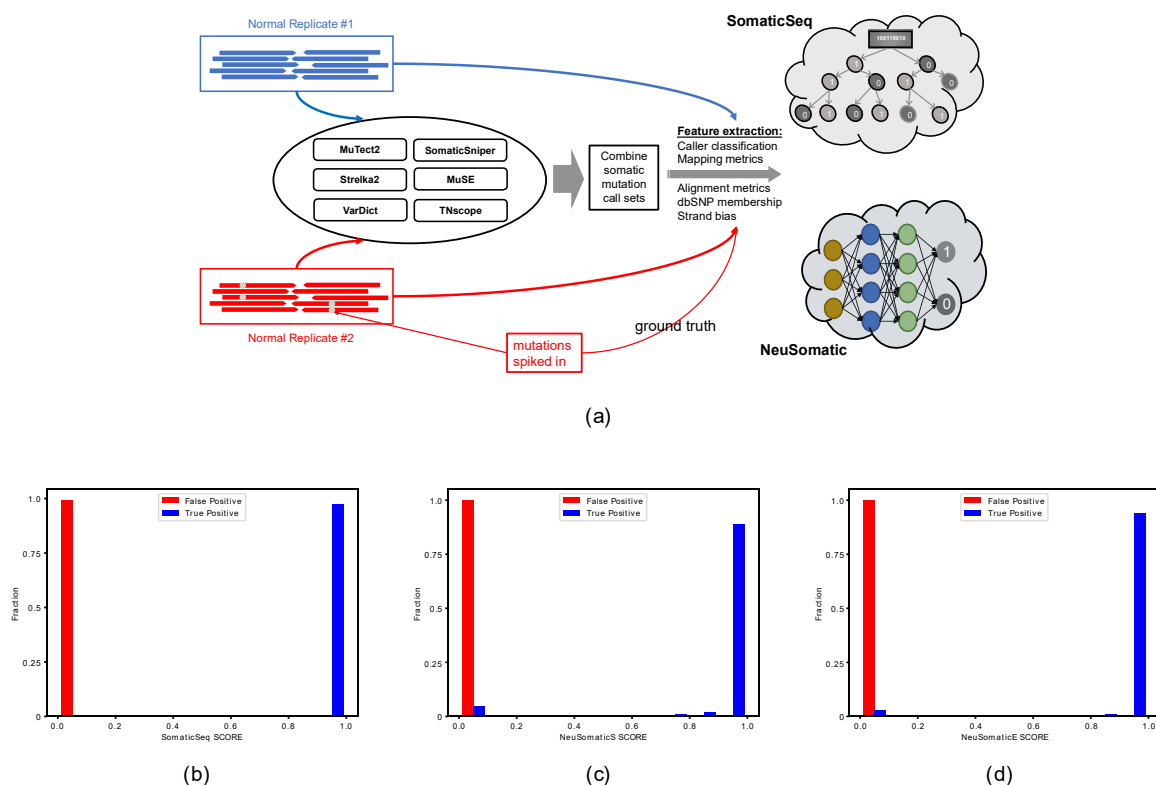
One-sided Z-tests were used to test the difference between two proportions, Pearson's correlation coefficient to calculate the correlation. A linear mixed model was applied to validate somatic mutations with tumor titration series and a generalized linear mixed model was specified to calculate the SNV/indel calling probability (SCP), both of which were detailed at the online method section. One-sided Z-tests and Pearson's correlation were calculated by python with the scipy package. The linear mixed model and generalized linear mixed model were fitted in R.

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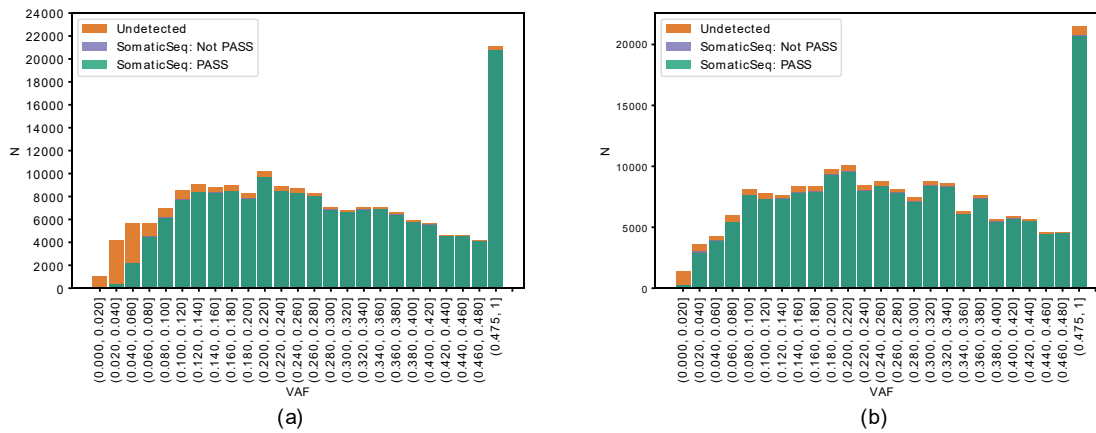
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https://www.accessdata.fda.gov/cdrh_docs/pdf17/P170019C.pdf.

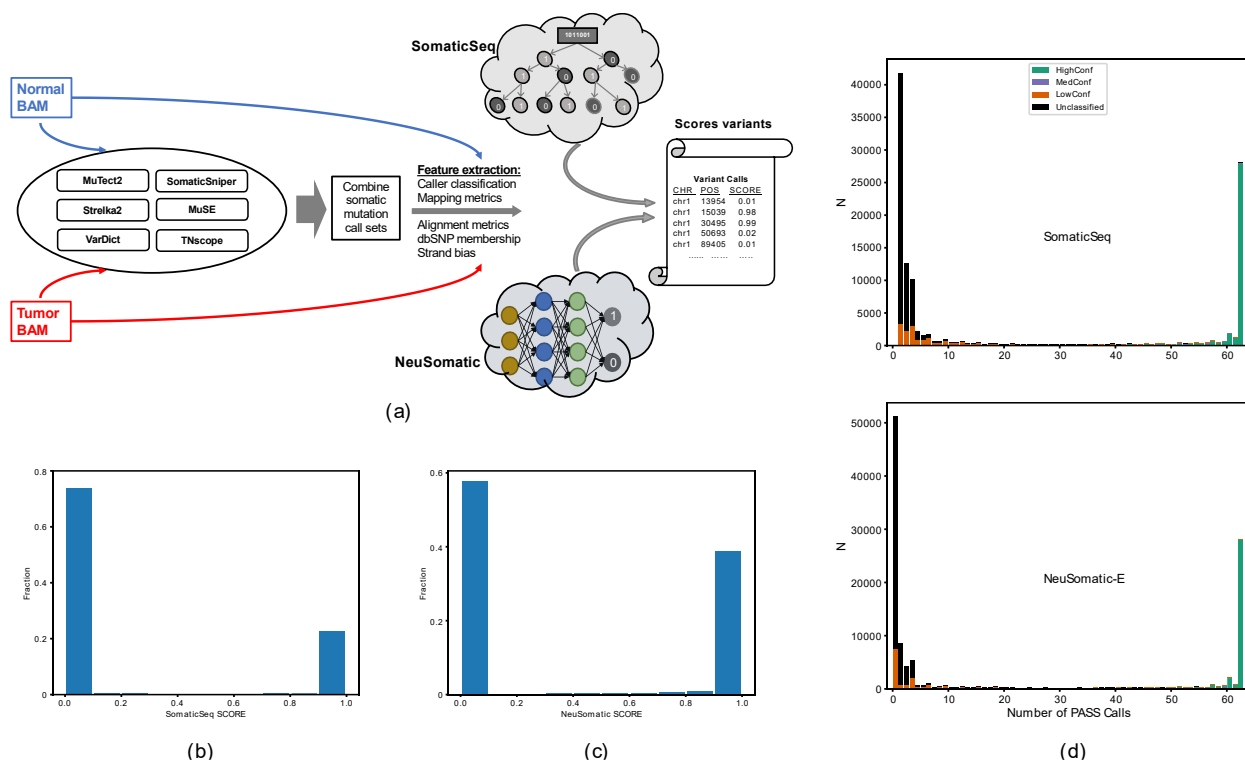
Supplementary Figures



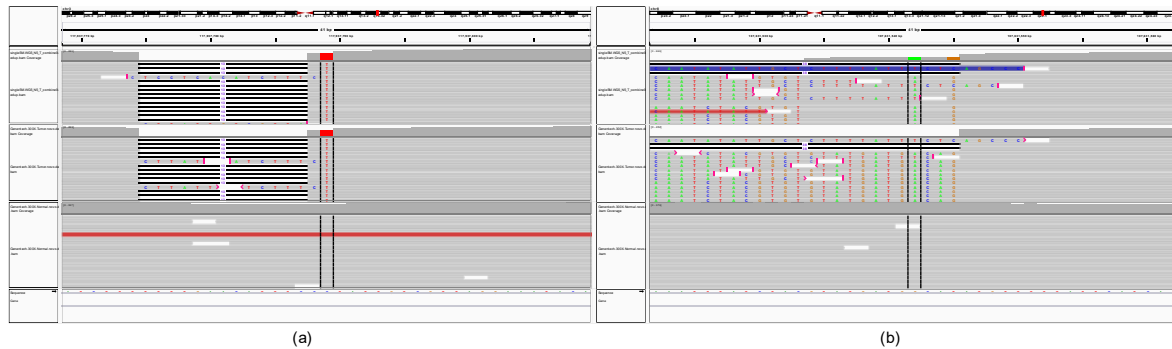
Supplementary Figure 1) Using simulated data to build the classifier for machine learning models: SomaticSeq and NeuSomatic. **(a)** The schematic of training workflow to create machine learning classifiers. **(b)** Histogram of SomaticSeq scores assigned to true positives and false positives. Most calls are either > 0.9 (likely true positive) or < 0.1 (likely false positive), with only a tiny fraction of calls in between (1.03%). **(c)** Histogram of NeuSomatic-E scores. **(d)** Histogram of NeuSomatic-S scores.



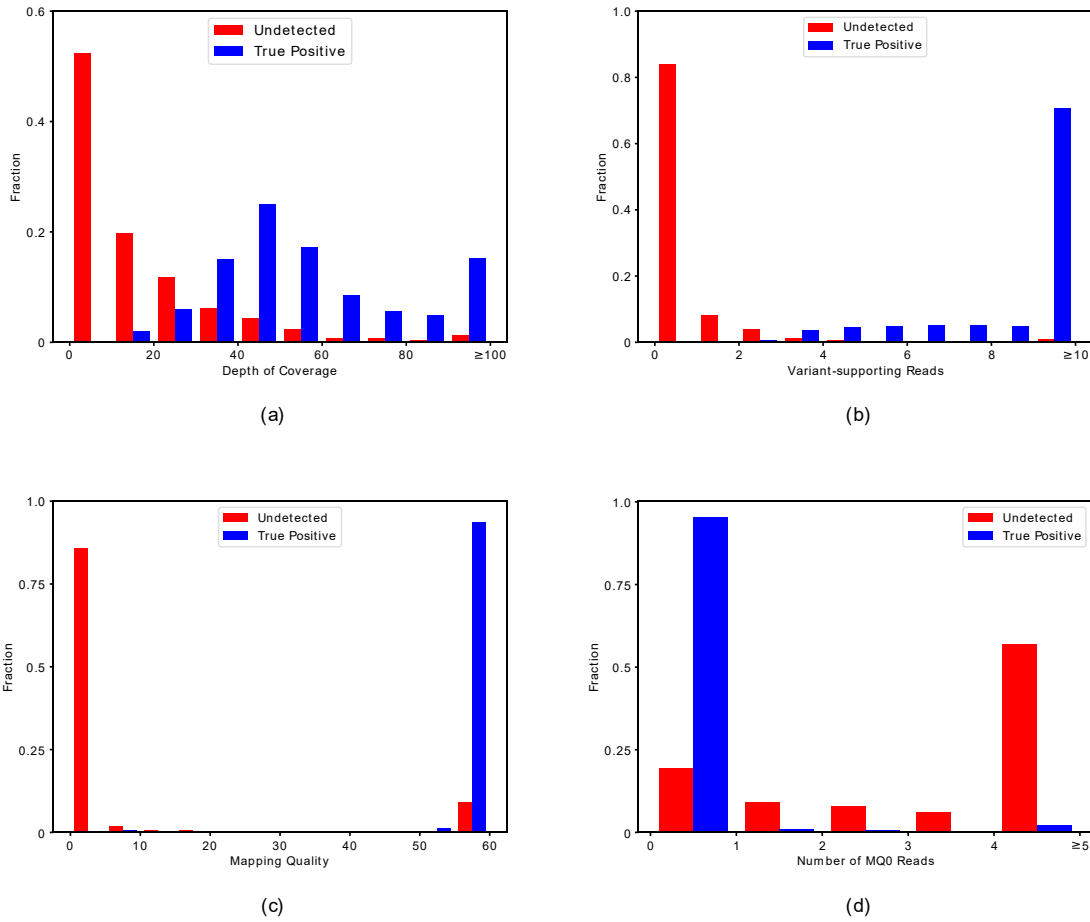
Supplementary Figure 2) Histogram of BAMSurgeon simulated variant. **(a)** Training data sets with approximately 50X in sequencing depth. **(b)** Training data set with 203X tumor depth and 160X normal depth. The green bars represent the variants that were classified correctly as true positives by SomaticSeq. The purple bars (on top of green bars), hardly visible in many cases, represent true positives that were detected by at least one caller but were not correctly classified as true positive (PASS) by SomaticSeq. The orange bars represent variants that were not detected by any of the six callers.



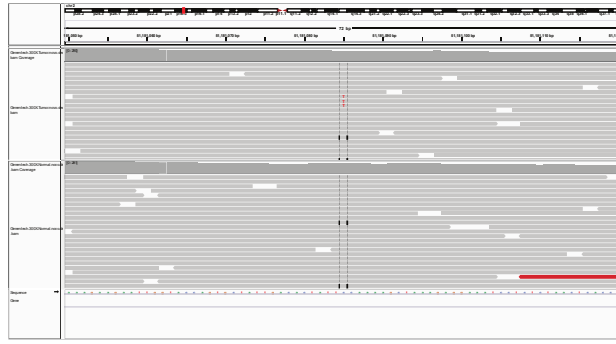
Supplementary Figure 3) Using SomaticSeq and NeuSomatic to classify somatic mutation calls of each pair of BAM files (a) Mutation calling workflow for each pair of tumor-normal BAM files. The SomaticSeq classifiers are specific to each sequencing center and each aligner as depicted in **Suppl. Fig. 1a. NeuSomatic classifiers were trained on combined training data sets. (b) Histogram of SomaticSeq scores for our call sets. This is qualitatively similar to the cross-validated histogram shown in **Suppl. Fig. 1b**. (c) Histogram of NeuSomatic-E scores for our call sets. (d) The number of tumor-normal data sets (out of 63 in total) where SomaticSeq (top) and NeuSomatic-E (bottom) have called a variant PASS. It is apparent that the distribution is bimodal, i.e., HighConf calls have been classified as PASS by both SomaticSeq and NeuSomatic-E in the vast majority of 63 data sets, whereas Unclassified calls have been classified as PASS only a few times.**



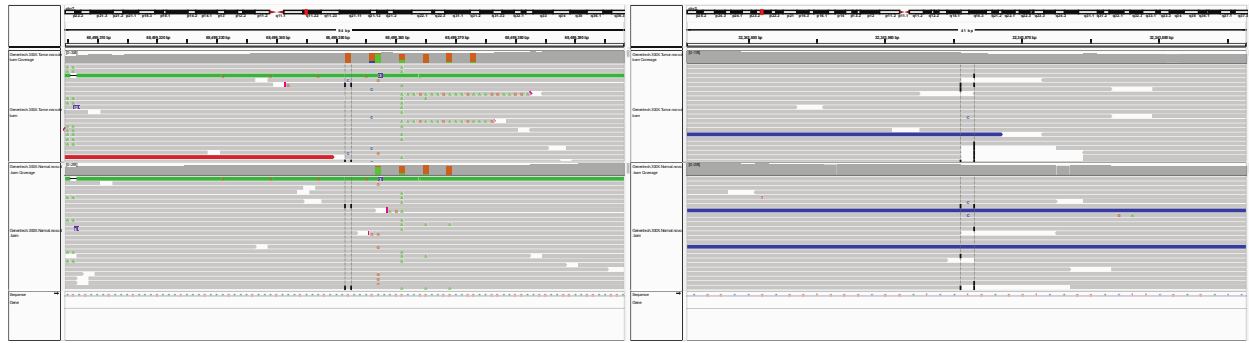
Supplementary Figure 4) Integrative Genome Viewer display of calls of interest. For each of the sub figures, the top alignments are NovaSeq tumor aligned with BWA. The middle alignments are HiSeq tumor aligned with NovoAlign. The bottom alignments are HiSeq normal aligned with NovoAlign. **(a)** The chr3:117937789 C>T is 1-bp away from the 13-bp deletion, which brings into question the mapping/alignment. It may be a true complex variant. **(b)** The chr8: 107631542 G>A that seemed to be in the middle of a 38-dp deletion. This looks like a false positive even though it was deemed “validated” by our rules; it may be correctly labeled as “Unclassified.”



Supplementary Figure 5) Undetected high-VAF in silico mutations. **(a)** Depth of coverage for undetected vs. correctly classified simulated variants. For correctly classified variants, the depth of coverage peaked around 50X, which was our target coverage. For the undetected variants; however, the highest peak represented depth of coverage of less than 10. **(b)** Number of variant-supporting reads. Many of the undetected variants had zero variant-supporting reads, due to low-MQ mutated reads being mapped to a different location after in silico mutation. **(c)** Most of the undetected variants have BWA mapping quality scores below 5. **(d)** The majority of undetected variants had at least five MQ=0 reads mapped to that position, indicating low mappability regions.



(a)



(b)

(c)

Supplementary Figure 6) IGV screenshots of a false negative and two false positives that are representative of the 20 visually inspected calls in our benchmarking of Strelka2 called on NV_T_2 vs. NV_N_2 with NovoAlign. The top alignments represent pure tumor of a 350X data. The bottom alignments represent pure normal of a 350X data. **(a)** A false negative at chr2:51181085 C>T, where there was only one variant read in NT_V_2 so it was not called. In the 350X data, however, there were 9 variant reads with a VAF of 4%. **(b)** A false positive at chr7:68499352 in a GAAA tandem repeat region with numerous read-to-reference mismatches everywhere in the region. This looks like a clear false positive. **(c)** A false positive at chr2:32343566. Both tumor **and** normal have 5 variant-supporting reads in the 350X data. There were no variant-supporting reads in NV_N_2, so it was called a somatic mutation in that replicate.

PASS = 1	LowQual = 0	REJECT = -1	SUM	Group Score
3	0	0	3	3
2	1	0	2	
2	0	1	1	1
1	2	0	1	
1	1	1	0	0
0	3	0	0	
1	0	2	-1	
0	2	1	-1	
0	1	2	-2	-3
0	0	3	-3	

9	0	0	9	3
8	1	0	8	
8	0	1	7	
7	2	0	7	
7	1	1	6	
6	3	0	6	
7	0	2	5	
6	2	1	5	
5	4	0	5	
6	1	2	4	
5	3	1	4	
4	5	0	4	
6	0	3	3	1
5	2	2	3	
4	4	1	3	
3	6	0	3	
5	1	3	2	
4	3	2	2	
3	5	1	2	
2	7	0	2	
5	0	4	1	
4	2	3	1	
3	4	2	1	
2	6	1	1	0
1	8	0	1	
4	1	4	0	
3	3	3	0	
2	5	2	0	
1	7	1	0	
0	9	0	0	
4	0	5	-1	
3	2	4	-1	
2	4	3	-1	
1	6	2	-1	
0	8	1	-1	-3
3	1	5	-2	
2	3	4	-2	
1	5	3	-2	
0	7	2	-2	
3	0	6	-3	
2	2	5	-3	
1	4	4	-3	
0	6	3	-3	
2	1	6	-4	
1	3	5	-4	
0	5	4	-4	
2	0	7	-5	
1	2	6	-5	
0	4	5	-5	
1	1	7	-6	
0	3	6	-6	
1	0	8	-7	
0	2	7	-7	
0	1	8	-8	
0	0	9	-9	

# of Score=3's	Group Score #1	Group Score #2	Group Score #3	Group Score #4	Group Score #5	SUM	Aligner-Centric Classification
5	3	3	3	3	3	15	Strong Evidence
4	3	3	3	3	1	13	
4	3	3	3	3	0	12	
3	3	3	3	1	1	11	
3	3	3	3	1	0	10	
4	3	3	3	3	-3	9	
3	3	3	3	0	0	9	
2	3	3	1	1	1	9	
2	3	3	1	1	0	8	
3	3	3	3	1	-3	7	
2	3	3	1	0	0	7	
1	3	1	1	1	1	7	
3	3	3	3	0	-3	6	Weak Evidence
2	3	3	0	0	0	6	
1	3	1	1	1	0	6	
2	3	3	1	1	-3	5	
1	3	1	1	0	0	5	
0	1	1	1	1	1	5	
2	3	3	1	0	-3	4	
1	3	1	0	0	0	4	
0	1	1	1	1	0	4	
3	3	3	3	-3	-3	3	
2	3	3	0	0	-3	3	
1	3	1	1	1	-3	3	Neutral Evidence
1	3	0	0	0	0	3	
0	1	1	1	0	0	3	
1	3	1	1	0	-3	2	
0	1	1	0	0	0	2	
2	3	3	1	-3	-3	1	
1	3	1	0	0	-3	1	
0	1	1	1	1	-3	1	
0	1	0	0	0	0	1	
2	3	3	0	-3	-3	0	
1	3	0	0	0	-3	0	
0	1	1	1	0	-3	0	Likely False Positive
0	0	0	0	0	0	0	
1	3	1	1	-3	-3	-1	
0	1	1	0	0	-3	-1	
1	3	1	0	-3	-3	-2	
0	1	0	0	0	-3	-2	
2	3	3	-3	-3	-3	-3	
1	3	0	0	-3	-3	-3	
0	1	1	1	-3	-3	-3	
0	0	0	0	0	-3	-3	
0	1	1	0	-3	-3	-4	
1	3	1	-3	-3	-3	-5	
0	1	0	0	-3	-3	-5	
1	3	0	-3	-3	-3	-6	
0	0	0	0	-3	-3	-6	
0	1	1	-3	-3	-3	-7	
0	1	0	-3	-3	-3	-8	
1	3	-3	-3	-3	-3	-9	
0	0	0	-3	-3	-3	-9	
0	1	-3	-3	-3	-3	-11	
0	0	-3	-3	-3	-3	-12	
0	-3	-3	-3	-3	-3	-15	

Supplementary Figure 7) Intermediate Scoring system. The upper left table depicts all the different combinations of per sample classification when a Data Group has 3 replicates: PASS = +1, REJECT = -1, and 0 otherwise. Their sum was converted to four possible Group Scores of 3, 1, 0, and -3. The lower left table depicts the same thing as the upper left, except it is for Data Groups with 9 replicates. The table on the right depicts the different combinations of the five Group Scores that led to the four aligner-centric classifications for each aligner.

Supplementary Tables

Pre-final label	Number	≥ 1 aligner	≥ 2 aligners
HighConf	37456	99.6%	97.2%
MedConf	1291	93.0%	77.0%
LowConf	15584	24.0%	13.6%
Unclassified	37901	1.7%	0.4%

Supplementary Table 1) This table shows the number of variants in each confidence level and their concordance with the 1,500X NeuSomatic call sets before the confidence labels were further polished. The ≥ 1 aligner column shows the percentage of variant calls that were rediscovered in at least one of the three 1,500X call sets (one call set for each of the three aligners). The ≥ 2 aligners column counts a variant as concordant if it is called in at least two out of three 1,500X call sets.

Platforms	Total Calls	Substitution Errors	Indel Errors	Substitution Error Rate	Indel Error Rate
HiSeq	1,818,768,592,271	2,081,962,721	35,530,669	0.114%	0.002%
NovaSeq	1,009,625,000,339	860,881,365	24,258,829	0.085%	0.002%
PacBio	121,639,927,836	4,464,346,903	8,596,779,700	3.670%	7.067%

Supplementary Table 2) The error rates were estimated as follows: in HCC1395BL sequencing data, every genomic position in the high-confidence region, except for the germline variant positions, is assumed to be a reference base. Error rates were calculated based on the putative reference bases only. In aligned reads (i.e., BAM files), every non-reference base call covering the putative reference base position is assumed to be an error and counted as such. HiSeq and NovaSeq have very similar error profiles, i.e., approximately one substitution error for every 1000 bases and one indel error for every 50,000 bases. The error rate of PacBio long-read sequencing data is higher than the detection limit of our somatic mutation reference call set.

Validation Platform	Variant Type	Category	Total Number	Fraction Interpretable	Validation Rate (Interpretable)	Validation Rate (Total)
AmpliSeq Deep Sequencing	SNV	SCP = 1	422	(419/422) 99.3%	(416/419) 99.3%	98.6%
		1 > SCP ≥ 0.9	63	(54/63) 85.7%	(41/54) 75.9%	65.1%
		0.9 > SCP ≥ 0.5	68	(56/68) 82.4%	(17/56) 30.4%	25.0%
		0.5 > SCP > 0	11	(11/11) 100.0%	(0/11) 0.0%	0.0%
	INDEL	SCP = 1	48	(47/48) 97.9%	(46/47) 97.9%	95.8%
		1 > SCP ≥ 0.9	23	(20/23) 87.0%	(19/20) 95.0%	82.6%
Ion Torrent WES	SNV	SCP = 1	44451	(43148/44451) 97.1%	(42052/43148) 97.5%	94.60%
		0.5 > SCP > 0	4	(4/4) 100%	(0/4) 0%	0%
	INDEL	SCP = 1	2899	(2625/2899) 90.5%	(2536/2625) 96.6%	87.50%

Supplementary Table 3) AmpliSeq and Ion Torrent WES validation rate for germline variants.

Category	Callable		Non-Callable	
	Number	Fraction	Number	Fraction
HighConf	37,498	40.65%	0	0.00%
MedConf	2,162	2.34%	336	1.28%
LowConf	14,901	16.15%	3,896	14.80%
Unclassified	37,787	40.96%	22,086	83.92%
Total	92,248	100.00%	26,318	100.00%

Supplementary Table 4) The number of SNV calls of each category in the consensus callable regions and those outside of them. Consensus callable regions made up 92% of the entire genome.

Panels	Region size	Total SNVs	Total non-syn SNVs	95% interval of Inferred SNV/Mbp	95% interval of Inferred non-syn SNV/Mbp
SEQ-II high-confidence regions	2,485,467,324	39,560	---	15.9	---
UCSC Genecode v29 Coding	31,101,689	473	352	14.6 (13.2 – 15.8)	10.9
TruSight One Expanded v2.0	14,607,935	209	141	14.3 (12.4 – 16.4)	9.7 (8.1 – 11.4)
TruSight One v1.1	10,542,342	147	100	13.9 (11.8 – 16.4)	9.5 (7.7 – 11.5)
MSK-IMPACT ¹⁴	934,044	9	7	9.6 (4.4 – 18.3)	7.5 (3.0 – 15.4)
309 cancer-related gene coding regions*	715,933	10	7	14.0 (6.7 -- 25.7)	9.8 (3.9 - 20.1)
TruSight Tumor 170	505,593	8	5	15.8 (6.8 – 31.2)	9.9 (3.2 – 23.0)
NEB Next Direct Cancer HotSpot	35,392	2	2	56.5 (6.8 – 204.1)	56.5 (6.8 – 204.1)
AmpliSeq Cancer Hot Spot v2	21,048	2	2	95.0 (11.5 – 343.2)	95.0 (11.5 – 343.2)
TruSight Tumor 26	13,792	1	1	72.5 (17.6 – 523.7)	72.5 (17.6 – 523.7)

Supplementary Table 5) Tumor mutation burden (TMB) benchmarks on various panels. *These 309 genes are taken from FoundationOne CDx technical information¹⁵, but this is not the same panel as it does not include additional regions. The Region Sizes are calculated as the intersection between the region sizes and the SEQC-II high-confidence regions.

CHROM	POS	REF	ALT	VALIDATION	Comments
chr3	117937789	C	T	YES	One base away from a 13-bp deletion and filled with soft-clipped reads
chr8	107631542	G	A	YES	In the middle of a 38-bp deletion, and the end of AmpliSeq coverage
chr8	107631545	T	G	YES	In the middle of a 38-bp deletion, and the end of AmpliSeq coverage
chr11	24722179	T	G	YES	One base away from a 13-bp deletion and filled with soft-clipped reads

Supplementary Table 6) The four Unclassified SNV calls that were “validated” by AmpliSeq

CHR	POS	VAF	REF	ALT	Category	AmpliSeq					Ion Torrent				
						Validation	Tumor VDP	Tumor DP	Normal VDP	Normal DP	Validation	Tumor VDP	Tumor DP	Normal VDP	Normal DP
chr9	105607844	0.669	G	A	HighConf	Confirmed	1971	4083	2	5611	Confirmed	32	59	0	65
chr9	129102404	0.229	T	C	HighConf	Confirmed	321	1301	0	1096	Confirmed	3	13	0	28
chr12	131844177	0.997	G	C	HighConf	Confirmed	390	396	0	506	Confirmed	28	28	0	58
chr15	28815512	0.471	G	A	HighConf	Confirmed	771	1589	0	1975	Not Confirmed	0	46	0	82
chr17	4545725	0.356	T	C	HighConf	Confirmed	2037	6394	5	5652	Confirmed	3	18	0	38
chr19	35488730	0.506	A	C	LowConf	Confirmed	2642	5087	0	4124	Confirmed	41	83	0	91
chr22	23875150	0.05	G	A	MedConf	Confirmed	200	4832	0	2596	Confirmed	3	82	0	71

Supplementary Table 7) The seven variant calls that were covered by both AmpliSeq Targeted Sequencing and Ion Torrent WES. There was no variant-supporting read for chr15:28815512 in Ion Torrent because none of the 25 variant-supporting reads met the minimum MQ threshold but this was also validated by AmpliSeq.

	SNV		INDEL	
	Truth Set Classification	Cross Validation	Truth Set Classification	Cross Validation
Sensitivity	98.39%	99.21%	93.79%	92.11%
Specificity	99.19%	99.67%	94.64%	97.00%
PPV	99.63%	99.74%	94.64%	96.53%
NPV	99.87%	99.91%	99.10%	98.37%

Supplementary Table 8) Accuracy of the classification for the 203X synthetic tumor vs. 160X normal. Truth set classification denotes the classification accuracies when the variant calls were classified by the classifiers created for single replicates, so their sequencing depth is about 1/4 the test data. Cross validation refers to classifiers for two equivalent training sets, so their sequencing depths are very similar.

Categories	Initial Tiers	Extra Criteria	SNV	INDEL	SNV	INDEL
Tier 1	AllPASS	All 63 data sets are classified as PASS	26,524	421	36,613	1,340
All three Aligner-Centric Classifications are Strong Evidence	Tier 1A	For each Aligner-Centric Classification, each of the 5 Data Groups are deemed “+3”	4,599	467		
	Tier 1B	For each Aligner-Centric Classification, $\geq 3/5$ of the Data Groups are deemed “+3”	4,844	391		
	Tier 1C	No extra criterion	646	61		
Tier 2	Tier 2A	For the 3rd Aligner-Centric Classification that’s not Strong, it is classified as Weak	362	86	543	497
Two out of three Aligner-Centric Classifications are Strong Evidence	Tier 2B	The 3rd Aligner-Centric Classification is Neutral.	73	30		
	Tier 2C	No extra criterion: the 3rd Aligner-Centric Classification is deemed Likely False Positive.	108	381		
Tier 3	Tier 3A	The 2 other Aligner-Centric Classification are Weak	125	12	373	133
One out of three Aligner-Centric Classifications is Strong Evidence	Tier 3B	1 of the other 2 Aligner-Centric Classification is Weak	95	60		
	Tier 3C	No extra criterion: none of the other 2 is even Weak	153	60		
Tier 4	Tier 4A	All 3 Aligner-Centric Classifications are Weak	692	40	54,149	3,028
None of the three Aligner-Centric Classifications is Strong Evidence	Tier 4B	2/3 of the Aligner-Centric Classifications are Weak	276	43		
	Tier 4C	1/3 of the Aligner-Centric Classifications are Weak	252	107		
	REJECT	All Aligner-Centric Classification is Neutral / Likely False Positive	52,929	2,838		

Supplementary Table 9) The criteria for each tier assignment, as well as the number of variant calls that fell into each category. Most calls fell either into Tier 1 or Tier 4, which means these calls either had broad evidence to support their somatic status, or very little evidence at all.

Software	Version	Sources
BAMSurgeon	v1.1	https://github.com/adamewing/bamsurgeon
Bowtie2	v2.2.9	http://bowtie-bio.sourceforge.net/bowtie2
BWA MEM	v0.7.15	https://github.com/lh3/bwa
DeepVariants	v0.5.1	https://github.com/google/deepvariant
Freebayes	v1.2.0	https://github.com/ekg/freebayes
GATK3	v3.8	https://software.broadinstitute.org/gatk/download/archive
hap.py	V0.3.10	https://github.com/Illumina/hap.py
HaplotypeCaller	v3.8.0	https://software.broadinstitute.org/gatk/download/archive
MuSE	v1.0rc_c	https://github.com/danielfan/MuSE
MuTect2	4.beta.6	https://software.broadinstitute.org/gatk/gatk4
NeuSomatic	v0.1.3	https://github.com/bioinform/neusomatic
NovoAlign	v3.07.01	http://www.novocraft.com/products/novoalign/
RTG	V3.8.4	https://github.com/RealTimeGenomics/rtg-tools
SomaticSeq	v2.8.1 / seqc2_v1.2	https://github.com/bioinform/somaticseq
SomaticSniper	v1.0.5.0	https://github.com/genome/somatic-sniper
Strelka2	v2.8.4	https://github.com/Illumina/strelka
TMAP	v3.0.1	https://github.com/iontorrent/TS/tree/master/Analysis/TMAP
TNScope	v201711.02	https://www.sentieon.com/products/#tnscope
Trimmomatic	v0.30	http://www.usadellab.org/cms/index.php?page=trimmomatic
VarDict	v1.5.1	https://github.com/AstraZeneca-NGS/VarDictJava

Supplementary Table 10) Software, version, and download location